

Introductory Fluid Mechanics

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1 Introduction To Fluid Mechanics

This handout is made to outline some of the main topics covered in MIE312: Fluid Mechanics I, at University of Toronto. Congratulations for making it all the way to third year! Fluid mechanics is a beautiful subject, however, it could be rather difficult if we are not confident in our fundamental knowledge of some basic differential equations and multivariable calculus for mathematics, and basic Dynamics and Thermodynamics for our physical intuition. As a matter of fact, I will spend the first chapter to simply go over some thermodynamics, as it is fundamental to fluid mechanics.

With that said, bear in mind that fluid mechanics is slightly different from the other courses you may have had so far. We've had plenty of experience working with solids objects, but fluids are typically overlooked in a high school physics curriculum or first year university physics. So you may struggle at first, and the key is to always think of a simple example. If you do not understand a topic, just imagine some fluid flowing through a pipe. That is the most basic possible flow and it will guide your intuition through all the concepts we will cover.

2 Review of Thermodynamics

We may recall from our Thermodynamics class the ideal gas relation:

$$Pv = RT$$

Furthermore we have the first law of Thermodynamics which is a general statement of the conservation of energy:

$$dU = \delta W + \delta Q$$

So when we heat up an object at constant pressure, we do work by changing the volume, and if we heat up an object at constant volume we change the temperature of the object. These are all important equations for a control mass, where only energy can move in and out. For a control volume, we allow mass as well as energy, to move in and out. Thus, we start to talk about mass flow rate, $\dot{m} = \rho AV$, for which the first law states:

$$\dot{W} + \dot{Q} = \dot{m} \left(h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} + gz_2 - gz_1 \right)$$

For an isentropic flow, we have that $s_2 = s_1$, and thus,

$$\frac{dS}{dt} = \frac{\delta Q}{T} + \dot{m}(s_1 - s_2) + \dot{S}_{gen} = 0$$

where the first term is the entropy due to heat transfer, second term is entropy due to mass transfer, and the last term is the generated entropy. For isentropic flow, we then have:

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2} \right)^{\gamma-1} = \left(\frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$$

Where γ is the specific heat ratio $\gamma = C_p/C_v$.

The most important thing you need to take from this section is the first law, which will come back again when we derive the conservation of energy for a fluid. The isentropic flow relations will also be of use if you intend on continuing your studies in fluid mechanics, especially in the subject of compressible flows.

3 Fluid Properties and The Continuum Assumption

The word fluid, means generally any gas or liquid. So this course is really a class on mechanics of "not-solids". The main property we will see again and again in fluid mechanics is the density, ρ , so I will not extend the discussion on fluid properties too much. But we have to change how we think

about fluids. Typically, with solids we must think of the bulk of the material. With a fluid flow, you should think about layers of fluid that are sliding on top of each other, similar to having a stack of boxes slipping against each other. Each of these boxes moves at a slightly different velocity and there is friction between them. It is the exact same with fluids. This way of thinking about fluids is called the continuum assumption. Formally, the continuum assumption states that the velocity of a fluid can be described by a smooth vector field $\mathbf{u}(x, y, z, t)$. Furthermore, the density, pressure and temperature are also smooth scalar fields $T(x, y, z, t)$, $p(x, y, z, t)$ and $\rho(x, y, z, t)$. For this handout which covers MIE312, we will consider the density function to be constant. At higher temperatures and velocities, this assumption does not hold.

The continuum assumption is a rather nice assumption, however, we must be aware of situations where it does not hold. For example, in hypersonic flows where our velocities exceed 5 times the speed of sound, we can no longer think of a fluid as layers with continuous velocity and pressure fields, we have to start thinking about it as particles and that is where we need knowledge of statistical mechanics and specific flow equations for hypersonic speeds.

4 Viscosity and Shear

Other than density, there is a physical property that is specific to fluids which we have not yet discussed. That is called the viscosity. The viscosity, sometimes called the dynamic viscosity, μ , is a measure of a fluid's resistance to flow. The more viscous a fluid is, the harder it is to make it flow. A highly viscous fluid example is honey, and an example of a low viscosity fluid is water or ethanol.

If we go back to the analogy of a stack of boxes sliding against each other, the viscosity is analogous to the coefficient of friction between the boxes. Similarly, the shear, τ , of a fluid is analogous to the frictional force between the boxes. How can we quantify this, you may ask? The shear due to a fluid flow is proportional to the gradient of the velocity in the normal direction, similar to how a friction force is proportional to the normal force. The constant of proportionality then is the viscosity:

$$\tau = \mu \frac{du}{dy} \quad \text{or} \quad \tau = \mu \frac{\partial u}{\partial y}$$

So the driving force for shear stress in fluids is a gradient of velocity in the direction normal to the flow. In human language, if we got a fluid moving in direction x , then the distribution of velocity

along the y axis is what creates a shear stress.

Imagine water flowing through a pipe with some velocity gradient du_1/dy . Now, suppose we replace water with honey, such that we have a velocity gradient du_2/dy . What can we say about these velocity gradients knowing that honey is more viscous than water? Well, viscosity by definition we said is a measure of a fluid's resistance to flow. Suppose we have a linear shear stress profile. Since honey is more viscous, it is harder to push the layers of honey against each other, due to the friction being greater. Therefore, it is more difficult to create a velocity gradient. In general, more viscous fluids will have lower velocity gradients and an inviscid fluid will have infinite velocity gradient if the shear stress profile is non-uniform.

5 Visualization and Analysis of Fluid Flow

Before we can analyze a flow, we need a language to describe it. With a solid this is easy: a ball has a position, so we simply follow the ball as it moves. With a fluid there is no single ball to follow. There are infinitely many tiny fluid particles moving at the same time, and we have to decide how we want to keep track of them. There are two natural choices, and the difference between them is probably the most important idea of this chapter.

5.1 Two Ways of Watching a River

Imagine sitting in a small boat that drifts down a river. You follow one particular parcel of water wherever it goes, and you record its position $\mathbf{x}_p(t)$ and its velocity $\mathbf{u}_p = d\mathbf{x}_p/dt$. This is the **Lagrangian** description, and it is exactly how we treat particles in Dynamics. Now imagine that you stand on a bridge instead, and watch the water go by underneath you. You do not care which parcel of water is passing, you only record the velocity of whatever water is at the point below you, as a function of time. This is the **Eulerian** description, and it is precisely the velocity field $\mathbf{u}(x, y, z, t)$ that we introduced with the continuum assumption.

Almost all of fluid mechanics is written in the Eulerian way, because in engineering we usually care about what happens *at a location* (the pressure on a wing, the velocity at the exit of a nozzle) and not about the life story of one particular fluid particle. The catch is that Newton's second law is a statement about particles, not about points in space. So we need a way to translate the Lagrangian idea "the rate of change felt by a moving particle" into the Eulerian language of fields.

5.2 The Material Derivative

Let $\phi(x, y, z, t)$ be any field, for example the temperature of the river. Suppose the river is being warmed by the sun, and also suppose your boat is drifting downstream into warmer water. Your thermometer will read a change in temperature for two different reasons: the water everywhere is heating up in time, and you are moving to a place where the water is warmer. Let us make this precise. The particle follows the path $\mathbf{x}_p(t) = (x_p(t), y_p(t), z_p(t))$, so the temperature it experiences is $\phi(x_p(t), y_p(t), z_p(t), t)$. By the chain rule,

$$\left. \frac{d\phi}{dt} \right|_{\text{particle}} = \frac{\partial\phi}{\partial t} + \frac{\partial\phi}{\partial x} \frac{dx_p}{dt} + \frac{\partial\phi}{\partial y} \frac{dy_p}{dt} + \frac{\partial\phi}{\partial z} \frac{dz_p}{dt}.$$

But the particle moves with the fluid, so $dx_p/dt = u$, $dy_p/dt = v$ and $dz_p/dt = w$, where (u, v, w) are the components of $\mathbf{u}$. (From now on v is a velocity component, we will not need the specific volume again.) We give this derivative its own name, the **material derivative**:

$$\frac{D\phi}{Dt} = \frac{\partial\phi}{\partial t} + u \frac{\partial\phi}{\partial x} + v \frac{\partial\phi}{\partial y} + w \frac{\partial\phi}{\partial z} = \frac{\partial\phi}{\partial t} + (\mathbf{u} \cdot \nabla)\phi.$$

The first term is the *local* change, the sun heating up the whole river. The second term is the *convective* change, the boat moving into warmer water.

Now the important step. The acceleration of a fluid particle is the material derivative of its velocity:

$$\mathbf{a} = \frac{D\mathbf{u}}{Dt} = \frac{\partial\mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u}, \quad \text{e.g.} \quad a_x = \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z}.$$

Look carefully at the term $(\mathbf{u} \cdot \nabla)\mathbf{u}$. It is the velocity multiplied by its own derivative, so it is nonlinear. This single term is the reason why fluid mechanics is hard, and it is the reason turbulence exists. Keep an eye on it, it will come back in every chapter.

Example: steady flow that still accelerates. Consider a steady one-dimensional flow through a converging nozzle, with $u(x) = U_0(1 + x/L)$. Since the flow is steady, $\partial u/\partial t = 0$, and nothing at any fixed location ever changes with time. Yet

$$a_x = u \frac{\partial u}{\partial x} = \frac{U_0^2}{L} \left(1 + \frac{x}{L}\right) \neq 0.$$

The fluid particles accelerate even though the flow is steady, because each particle is being carried into a region where the fluid is faster. If you find this strange, think of a car driving on a road with a speed limit that increases along the road: nothing about the road changes in time, but the car still accelerates.

5.3 Streamlines, Pathlines and Streaklines

There are three different curves we can draw to visualize a flow, and it is very common to mix them up.

A **streamline** is a curve that is everywhere tangent to the velocity field *at one instant*. If we freeze time and draw little arrows of $\mathbf{u}$ everywhere, the streamlines are the curves that connect the arrows. Being tangent to $\mathbf{u}$ means that a small displacement $d\mathbf{x} = (dx, dy, dz)$ along the curve is parallel to $\mathbf{u}$, and therefore

$$\frac{dx}{u} = \frac{dy}{v} = \frac{dz}{w}.$$

A **pathline** is the trajectory of one fluid particle over time, i.e. the Lagrangian path, obtained by integrating $d\mathbf{x}_p/dt = \mathbf{u}(\mathbf{x}_p, t)$. A **streakline** is the curve formed by all the particles that have passed through one specific point, for instance the line of smoke leaving a chimney or the dye that you keep injecting into a river from a single needle.

If the flow is steady, then the velocity field never changes, so every particle that passes through a given point follows exactly the same path, and all three curves are identical. If the flow is unsteady they are different. The classic example is a garden hose that you wave from side to side. Every water particle leaves the nozzle and then travels in a straight line (a pathline), but the water you see is a wavy sheet (the streakline), and the instantaneous streamlines are different again.

Example: stagnation point flow. Consider the steady flow $u = kx$, $v = -ky$ with $k > 0$, which describes fluid approaching a wall along $y = 0$. The streamlines satisfy

$$\frac{dy}{dx} = \frac{v}{u} = -\frac{y}{x} \implies \frac{dy}{y} = -\frac{dx}{x} \implies xy = \text{constant}.$$

These are hyperbolas, as sketched below. Fluid comes down towards the wall, slows down and is squeezed sideways. The acceleration of a particle is $a_x = u \partial u / \partial x = k^2 x$ and $a_y = v \partial v / \partial y = k^2 y$, and at the origin both vanish: this is the *stagnation point*, where the fluid comes to rest.

Finally, notice that by definition the velocity is tangent to a streamline, so *no fluid ever crosses a*

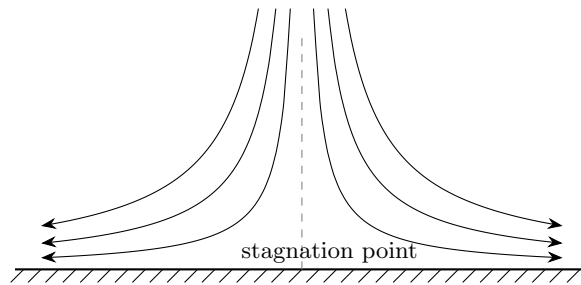


Figure 1: Streamlines $xy = \text{const}$ of the stagnation point flow $u = kx$, $v = -ky$.

streamline. If we draw a bundle of streamlines through a small closed curve we get a **streamtube**, which behaves like a pipe with invisible walls. Fluid that is inside the tube stays inside the tube. This is the idea behind one-dimensional flow analysis, and we will use it to derive Bernoulli's equation.

5.4 A Quick Vocabulary of Flow Types

Flows are classified in a few ways, and you should be comfortable with the words. A flow is *steady* if nothing depends on time and *unsteady* otherwise. It is *one-*, *two-* or *three-dimensional* depending on how many spatial coordinates the velocity depends on (the flow in a long straight pipe far from the inlet depends on the radius only, so it is one-dimensional). It is *internal* if it is confined by walls (pipes, ducts) and *external* if a body sits in an unbounded stream (a wing, a car). It is *viscous* or *inviscid* depending on whether friction matters, *laminar* or *turbulent* depending on whether the layers slide smoothly or mix chaotically, and *compressible* or *incompressible* depending on whether density changes matter. As we said, in this handout we will always take the density to be constant. Most of this course lives in steady, incompressible flows, and it is the viscous versus inviscid question that will split the rest of the handout in two.

6 Hydrostatics

Hydrostatics is the study of fluids at rest. It may sound like the boring chapter, but it is the easiest place to learn how pressure really works, and a lot of engineering (dams, tanks, ships, manometers,

the atmosphere itself) lives here. Recall from the viscosity chapter that shear stress needs a velocity gradient. A fluid at rest has no velocity gradients, hence no shear. So the only kind of stress a static fluid can carry is **pressure**, which always pushes perpendicular to a surface.

6.1 Pressure at a Point Acts Equally in All Directions

We can ask something that is not obvious: does the pressure at a point depend on the orientation of the surface we imagine there? Take a tiny wedge of fluid, of width dy into the page, with a vertical face of area $dz dy$, a horizontal face of area $dx dy$, and an inclined face of area $ds dy$, where $dx = ds \cos \theta$ and $dz = ds \sin \theta$. Let p_x , p_z and p_n be the pressures acting on these three faces. The outward normal of the inclined face is $(\sin \theta, \cos \theta)$, and pressure pushes *into* the wedge, so the force balance on the (static) wedge is

$$\begin{aligned} x : \quad p_x dz dy - p_n ds dy \sin \theta &= 0, \\ z : \quad p_z dx dy - p_n ds dy \cos \theta - \frac{1}{2} \rho g dx dz dy &= 0. \end{aligned}$$

Using $dz = ds \sin \theta$ in the first equation gives $p_x = p_n$. Using $dx = ds \cos \theta$ in the second gives $p_z - p_n = \frac{1}{2} \rho g dz$. Now shrink the wedge to a point, $dz \rightarrow 0$, and we find

$$p_x = p_z = p_n.$$

The physical reason is worth remembering. Pressure forces scale with an area (length^2), while the weight of the wedge scales with a volume (length^3). As the element shrinks, volume forces disappear faster than surface forces, so the surface forces have to balance among themselves. This is why pressure is a scalar field $p(x, y, z)$, and not a vector or a tensor.

6.2 The Hydrostatic Equation

Now let us find how pressure changes from point to point. Take a small cube of fluid with sides dx , dy , dz . The pressure on the left face is p , pushing to the right, and the pressure on the right face is $p + \frac{\partial p}{\partial x} dx$, pushing to the left. The net force in the x direction is

$$p dy dz - \left(p + \frac{\partial p}{\partial x} dx \right) dy dz = - \frac{\partial p}{\partial x} dx dy dz.$$

Doing the same in y and z , the net pressure force on the element per unit volume is $-\nabla p$. If the fluid is at rest, this must be balanced by the body force, which is the weight per unit volume $\rho \mathbf{g}$:

$$-\nabla p + \rho \mathbf{g} = 0.$$

Take z pointing up so that $\mathbf{g} = -g \hat{\mathbf{k}}$. The three components are

$$\frac{\partial p}{\partial x} = 0, \quad \frac{\partial p}{\partial y} = 0, \quad \frac{dp}{dz} = -\rho g.$$

In human language: in a static fluid the pressure changes only with height, and it decreases as you go up, because the higher you go the less fluid is sitting on top of you. It also tells us that surfaces of constant pressure are horizontal, which is why the surface of a lake is flat. For constant density, we integrate from a free surface at $z = z_0$ where the pressure is p_0 down to a depth $h = z_0 - z$:

$$p = p_0 + \rho gh.$$

The pressure at a point is just the pressure at the top plus the weight of the column of fluid above the point, per unit area. For water, ρgh for $h = 10$ m is about 98 kPa, i.e. roughly one extra atmosphere for every ten meters of depth. Engineers use *absolute pressure* (measured from a perfect vacuum) and *gauge pressure* (measured relative to the local atmosphere), $p_{gauge} = p_{abs} - p_{atm}$. Most of the time in fluid mechanics only pressure differences matter, so we will happily work with gauge pressure.

Compressible case: the atmosphere. When the density changes with height, we can no longer integrate directly, but we can use the ideal gas law $p = \rho RT$ from our thermodynamics review to write $\rho = p/RT$. Then $dp/dz = -\rho g$ becomes

$$\frac{dp}{p} = -\frac{g}{RT} dz.$$

If the temperature is constant (an isothermal atmosphere), integration gives $p = p_0 e^{-gz/RT}$, so the pressure decays exponentially with altitude. In the troposphere the temperature falls linearly,

$T = T_0 - Bz$ with $B \approx 6.5$ K/km. Then

$$\ln \frac{p}{p_0} = -\frac{g}{R} \int_0^z \frac{dz'}{T_0 - Bz'} = \frac{g}{RB} \ln \left(1 - \frac{Bz}{T_0} \right) \implies \frac{p}{p_0} = \left(1 - \frac{Bz}{T_0} \right)^{g/RB},$$

where the exponent g/RB is about 5.26 for air.

6.3 Manometers

A manometer is a tube of liquid used to measure pressure differences, and it is a direct application of $p = p_0 + \rho gh$. There are just two rules. First, in a connected column of the same fluid at rest, the pressure is the same at the same height. Second, as you move down through a fluid the pressure increases by $\rho g \Delta h$, and as you move up it decreases by $\rho g \Delta h$. So the method is simply to start at a point of known pressure, walk through the manometer adding and subtracting $\rho g \Delta h$ terms, and end at a point of known pressure.

Example. A pipe carrying water at pressure p_A is connected to a U-tube filled with mercury, with the other end open to the atmosphere. The mercury interface on the pipe side lies $h_1 = 0.30$ m below the pipe centerline, and the mercury surface on the open side is $h_2 = 0.50$ m above that interface. Starting at A , going down through the water, then up through the mercury to the atmosphere:

$$p_A + \rho_w g h_1 - \rho_m g h_2 = p_{atm} \implies p_{A,gauge} = \rho_m g h_2 - \rho_w g h_1.$$

With $\rho_m = 13600$ kg/m³ and $\rho_w = 1000$ kg/m³, this gives $p_{A,gauge} = 66.7 - 2.9 = 63.8$ kPa. Notice how the heavy mercury allows us to measure a large pressure with a small column, that is why it was chosen.

6.4 Hydrostatic Force on a Submerged Plane Surface

How hard does water push on a dam or on a gate? The pressure is not uniform, so the total force is the integral of the pressure over the area, and we also need to know *where* the resultant acts. Consider a flat plate submerged in a liquid, inclined at an angle θ to the free surface. Let y be the coordinate measured along the plate, starting from the line where the plane of the plate meets the free surface. A point at y is at a depth $h = y \sin \theta$. Working with gauge pressure, $p = \rho g y \sin \theta$,

and the force is

$$F = \int_A p \, dA = \rho g \sin \theta \int_A y \, dA = \rho g \sin \theta \, y_c A = \rho g h_c A = p_c A,$$

where we used the definition of the centroid, $\int y \, dA = y_c A$. So the total force is the pressure *at the centroid* times the area. That is a very convenient result. To find the line of action we balance moments about the free surface line: the moment of the resultant $F y_{cp}$ must equal the sum of the moments of all the small forces,

$$F y_{cp} = \int_A y p \, dA = \rho g \sin \theta \int_A y^2 \, dA = \rho g \sin \theta \, I_x,$$

where I_x is the second moment of area about the free surface line. Dividing by $F = \rho g \sin \theta \, y_c A$ and using the parallel axis theorem $I_x = I_c + A y_c^2$:

$$y_{cp} = \frac{I_x}{y_c A} = y_c + \frac{I_c}{y_c A}.$$

Since $I_c/(y_c A) > 0$, the **center of pressure** is always below the centroid. Intuitively, the pressure grows with depth, so the lower parts of the plate are pushed harder and pull the resultant downward.

Example: vertical rectangular gate. Take a gate of height H and width b , with its top edge at the free surface. Then $y_c = H/2$, $A = bH$, $I_c = bH^3/12$, so

$$F = \rho g \frac{H}{2} bH = \frac{1}{2} \rho g b H^2, \quad y_{cp} = \frac{H}{2} + \frac{bH^3/12}{(H/2)bH} = \frac{2H}{3}.$$

Of course, this is nothing more than the fact that the pressure distribution is a triangle, its area is $\frac{1}{2}(\rho g H)H$ per unit width, and the resultant passes through the centroid of that triangle at $2/3$ of the depth. Sketching the pressure triangle is often faster than doing the integral.

6.5 Hydrostatic Force on a Curved Surface

For a curved surface the direction of the pressure force changes from point to point, so the trick is to avoid integrating over the surface directly. Instead, we isolate a block of *fluid*, bounded on one side by the curved surface, and apply static equilibrium to it. The block is at rest, so all the forces on it must balance. The fluid block feels three kinds of forces: its own weight W , the pressure forces

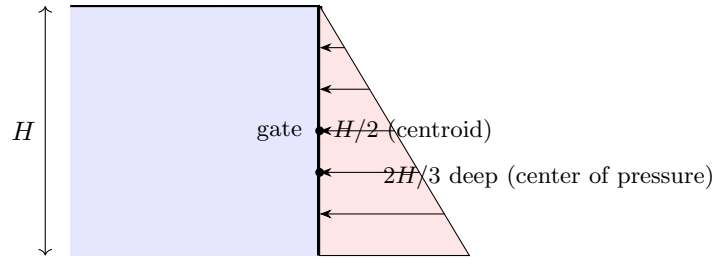


Figure 2: Triangular pressure distribution on a vertical gate. The resultant acts at the centroid of the triangle, which is $2H/3$ below the free surface.

from the rest of the fluid on its plane faces (which we can compute with the previous section), and the force from the curved surface, which by Newton's third law is equal and opposite to the force the fluid exerts on the surface. Working out the balance in each direction gives two rules:

- The **horizontal** component of the force on a curved surface equals the force on the vertical projection of that surface (a plane area, so we can use $F = p_c A$ and the center of pressure).
- The **vertical** component equals the weight of the fluid directly above the surface, up to the free surface (or to where the free surface would be, if the fluid is below the surface), and it acts through the centroid of that volume of fluid.

The two components are then added vectorially, and for a circular arc the resultant always passes through the center of the circle, because every pressure force is normal to the surface and therefore points along a radius.

7 Buoyancy and Stability

7.1 Archimedes' Principle

Everyone knows that things float, but why? Consider a body completely submerged in a fluid at rest. The pressure acts on its surface, and the net pressure force is the sum of all the little forces $-p \mathbf{n} dA$. We can compute this without any integral by a simple thought experiment. Take the body away, and replace it by fluid of the same shape. This ball of fluid is at rest, so it is in equilibrium: the pressure forces on its surface must balance its weight, meaning that the net pressure force is $\rho_f g V$ pointing upward, acting through the center of gravity of the ball. But the pressure field

around the ball does not know or care what is inside it. Hence the force on the real body is exactly the same:

$$F_B = \rho_f g V_{disp}.$$

The buoyant force equals the weight of the displaced fluid, and it acts through the centroid of the displaced volume, called the **center of buoyancy** B .

We can double check with a formal calculation. Take a vertical cylinder of cross-section A , with its top at depth h_1 and its bottom at depth h_2 . The pressures on the top and bottom faces are $p_0 + \rho_f g h_1$ and $p_0 + \rho_f g h_2$, and the side pressures cancel by symmetry. The net upward force is $(p_2 - p_1)A = \rho_f g (h_2 - h_1)A = \rho_f g V$. Any shape can be made of many thin vertical columns like this.

A floating body sits at the depth where buoyancy equals weight, $\rho_f g V_{sub} = \rho_b g V_{tot}$, so the fraction of the body that is submerged is

$$\frac{V_{sub}}{V_{tot}} = \frac{\rho_b}{\rho_f}.$$

For ice ($\rho \approx 917 \text{ kg/m}^3$) in seawater ($\rho \approx 1025 \text{ kg/m}^3$), about 89% of the iceberg is below the surface.

7.2 Stability of Submerged Bodies

Being in equilibrium is not enough, we also need the equilibrium to be *stable*: if the body is disturbed slightly, does it come back? For a fully submerged body the weight W acts at the center of gravity G and the buoyancy acts at B . If we rotate the body, both forces stay vertical, but the points G and B move together with the body. If G is below B , then after a small rotation the two forces form a couple that rotates the body back. If G is above B , the couple rotates the body further away, and it flips over. So a submerged body is stable if and only if G lies below B . This is why submarines carry their heavy equipment low, and ballast is placed at the bottom of the vessel.

7.3 Stability of Floating Bodies: the Metacentric Height

A floating body is more subtle. When it heels by a small angle θ , the shape of the submerged volume changes, and so does the location of B . Let us find how much.

Let the body be symmetric, with the waterplane (the cross-section at the water line) having area A_{wp}

and let x be the horizontal coordinate measured from the center of the waterplane, perpendicular to the axis of roll. When the body heels by θ , a wedge of extra volume goes under water on one side, and an identical wedge comes out of the water on the other side. A small piece of the waterplane at x gains a volume $dV = x\theta dA$ (a column of height $x\theta$, taking x positive for the side that goes down, and negative for the side that goes up). The center of buoyancy moves because of this transfer of volume, and by taking moments of the volume about the centerline,

$$V \overline{BB'} = \int x(x\theta) dA = \theta \int x^2 dA = \theta I_{wp},$$

where V is the displaced volume and I_{wp} is the second moment of area of the waterplane about the roll axis. Hence $\overline{BB'} = \theta I_{wp}/V$. Now, the buoyant force acts vertically through B' , and it crosses the (tilted) centerline of the body at a point M called the **metacenter**. For small angles, the geometry gives $\overline{BB'} = \overline{BM} \theta$, so

$$\overline{BM} = \frac{I_{wp}}{V}.$$

The distance from the center of gravity to the metacenter is the **metacentric height**, $\overline{GM} =$

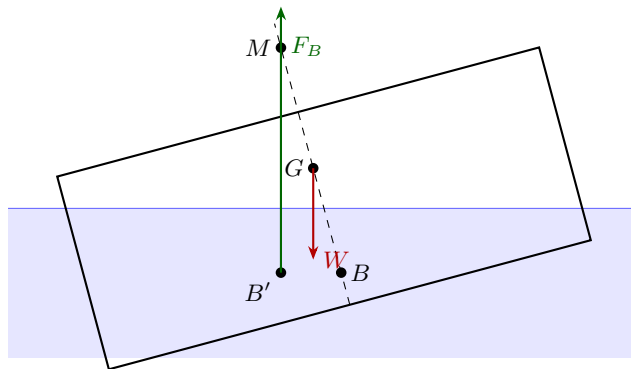


Figure 3: A heeled floating body. The buoyant force acts vertically through B' and crosses the body's centerline at the metacenter M . Here M is above G , so the couple formed by W and F_B pushes the body back upright.

$\overline{BM} - \overline{BG}$, where $\overline{BG}$ is positive when G is above B . When the body is heeled, the weight W acts through G and the buoyancy $F_B = W$ acts vertically through B' , so they form a couple of moment

$$M_{restoring} = W \overline{GM} \sin \theta.$$

If M lies above G ($\overline{GM} > 0$) the couple brings the body back to upright and the body is stable. If M lies below G the couple tips the body further over and the body capsizes.

Two consequences are worth remembering. First, $\overline{BM} = I_{wp}/V$ depends on the *shape*: a wide boat has a large I_{wp} (recall $I \sim bh^3/12$, so the width counts to the third power) and is therefore very stable in roll, while a narrow canoe is not. Second, loading a ship high up raises G , decreases $\overline{GM}$, and can make the ship unstable, which is why cargo and ballast are loaded as low as possible.

8 Control Volume Analysis

So far we have described fluids using fields, which is very powerful but requires us to know the flow everywhere. Very often in engineering we do not care about the details of the flow inside a device, we only care about what happens at the inlets and outlets: how much mass goes in and out, what force the fluid exerts on a pipe bend, how much power a pump needs. For this we go back to the Lagrangian idea of a system (a fixed collection of matter) and translate our familiar laws for a system (conservation of mass, Newton's second law, the first law of thermodynamics) into statements about a fixed region in space, called a **control volume**. The tool that performs this translation is the Reynolds Transport Theorem, and once we have it, the rest of this chapter is just applying it to mass, momentum and energy in turn.

8.1 The Reynolds Transport Theorem

Let B be any extensive property of a system (mass, momentum, energy: something that depends on how much stuff you have), and let $\beta = dB/dm$ be the corresponding intensive property (per unit mass). Then $B_{sys} = \int_{sys} \beta \rho dV$.

Picture a system that at time t occupies exactly a control volume CV fixed in space. A short time dt later, the system has moved (since it is made of the same fluid particles) to a new position. Some of it may have left through the boundary of the CV , call this region CS_{out} , and some new fluid may have flowed in, but let us focus on outflow for now.

By definition,

$$\frac{dB_{sys}}{dt} = \lim_{dt \rightarrow 0} \frac{B_{sys}(t + dt) - B_{sys}(t)}{dt}.$$

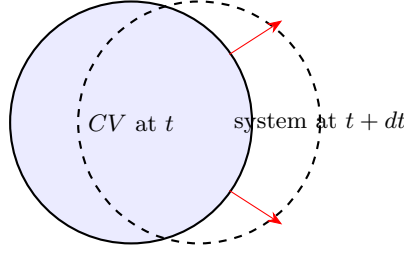


Figure 4: The system coincides with the CV at time t , and moves outward through part of the control surface by $t + dt$.

Now split $B_{sys}(t + dt)$: it consists of whatever is still inside the CV at $t + dt$, plus whatever has left through the boundary. And $B_{sys}(t)$ is, by our setup, exactly $B_{CV}(t)$. So

$$B_{sys}(t + dt) - B_{sys}(t) = [B_{CV}(t + dt) - B_{CV}(t)] + B_{out}(t + dt).$$

The first bracket, divided by dt , becomes $\partial/\partial t \int_{CV} \beta \rho dV$ in the limit: the rate of change of B stored inside the (fixed) control volume. For the second piece, consider a small patch dA of the control surface with outward normal $\mathbf{n}$. In time dt , the fluid crossing it sweeps out a little slanted cylinder of volume $(\mathbf{u} \cdot \mathbf{n}) dt dA$ (only the component of velocity normal to the surface carries fluid across it; the tangential component just slides along the surface). The amount of B carried across is $\beta \rho (\mathbf{u} \cdot \mathbf{n}) dt dA$. Summing over the whole control surface and dividing by dt :

$$\frac{dB_{sys}}{dt} = \frac{\partial}{\partial t} \int_{CV} \beta \rho dV + \oint_{CS} \beta \rho (\mathbf{u} \cdot \mathbf{n}) dA.$$

This is the **Reynolds Transport Theorem (RTT)**. In words: *the rate of change of B for a system equals the rate of change of B stored inside the control volume, plus the net rate at which B is carried out through the control surface by the flow.* The surface integral is often called the net *flux* term, and notice that where fluid flows in, $\mathbf{u} \cdot \mathbf{n} < 0$ (since $\mathbf{n}$ points outward), so inflow contributes negatively and outflow positively, exactly as it should.

The RTT is not itself a law of physics, it is a bookkeeping identity, a change of accounting from following matter to watching a fixed region. All the physics is still hidden in dB_{sys}/dt , which is where we plug in mass conservation, Newton's second law, or the first law, one at a time.

8.2 Conservation of Mass: Continuity

Take $B = m$, so $\beta = 1$. Since mass cannot be created or destroyed for a system, $dm_{sys}/dt = 0$, and the RTT gives immediately

$$\frac{\partial}{\partial t} \int_{CV} \rho dV + \oint_{CS} \rho(\mathbf{u} \cdot \mathbf{n}) dA = 0.$$

This is the **integral continuity equation**: the rate at which mass accumulates inside the CV plus the net mass flow rate leaving through the surface must equal zero, i.e. whatever mass builds up inside must be exactly the net inflow. For a control volume with a number of well-defined inlets and outlets over which the flow is roughly uniform, the surface integral collapses to a sum, $\oint \rho(\mathbf{u} \cdot \mathbf{n}) dA = \sum_{out} \dot{m} - \sum_{in} \dot{m}$, so for steady flow,

$$\sum \dot{m}_{out} = \sum \dot{m}_{in}, \quad \dot{m} = \rho AV.$$

This is exactly the mass flow relation we already used in the thermodynamics chapter, and now you can see where it comes from. For an incompressible flow through a single streamtube of varying area, ρ is constant and cancels, leaving $A_1 V_1 = A_2 V_2$: the flow speeds up where the tube narrows, precisely because the same volume of fluid has to pass every cross-section per unit time.

8.3 Conservation of Linear Momentum

Now take $B = m\mathbf{u}$, so $\beta = \mathbf{u}$. Newton's second law for a system says $d(m\mathbf{u})_{sys}/dt = \sum \mathbf{F}$, the sum of all forces acting on the system (its weight, and the pressure and shear forces exerted on it by whatever surrounds it). The RTT then gives the **linear momentum equation**:

$$\sum \mathbf{F} = \frac{\partial}{\partial t} \int_{CV} \mathbf{u} \rho dV + \oint_{CS} \mathbf{u} \rho(\mathbf{u} \cdot \mathbf{n}) dA.$$

This looks intimidating but it is simply Newton's second law rewritten for a fixed region: the net force on whatever fluid is currently inside the CV equals the rate of change of momentum stored inside plus the net rate at which momentum is carried out by the flow. Since $\mathbf{u}$ is a vector, this is really three scalar equations, one per coordinate direction, and it is the workhorse for computing forces on pipe bends, nozzles, jets and blades, because it lets us find the force *without knowing the*

pressure distribution in detail inside the device, only at the inlets and outlets.

Example: force on a reducing pipe bend. Consider a horizontal pipe bend that turns steady, incompressible flow through 90° , with inlet area A_1 , velocity V_1 , gauge pressure p_1 along x , and outlet area A_2 , velocity V_2 , gauge pressure p_2 along y . Choose the *CV* to be the fluid inside the bend, cut by the inlet and outlet planes. The forces on this fluid are the pressures pushing at the two open faces, and the (unknown) force $\mathbf{R}$ exerted by the pipe walls on the fluid, which is what we want to find. Applying the momentum equation in x and y for steady flow (so the accumulation term vanishes):

$$\begin{aligned} x : \quad p_1 A_1 + R_x &= \dot{m} V_2 \cos 90^\circ - \dot{m} V_1 = -\dot{m} V_1, \\ y : \quad p_2 A_2 \cdot (-1) + R_y &= \dot{m} V_2 - 0, \end{aligned}$$

using continuity $\dot{m} = \rho A_1 V_1 = \rho A_2 V_2$ to identify the mass flow rate, and being careful with the sign of each momentum flux ($\mathbf{u} \cdot \mathbf{n}$ is negative at the inlet and positive at the outlet, and each $\mathbf{u}$ points in its own flow direction). Solving,

$$R_x = -p_1 A_1 - \dot{m} V_1, \quad R_y = \dot{m} V_2 + p_2 A_2.$$

The force the fluid exerts *on the pipe* is $-\mathbf{R}$ by Newton's third law. Notice that both the pressure and the momentum change contribute; this is exactly why bends and nozzles need to be anchored, the flow is trying to push its way straight and the pipe wall has to redirect it, at the price of a reaction force.

8.4 Conservation of Energy and Bernoulli's Equation

For the last application, take β to be energy per unit mass, $e = \hat{u} + V^2/2 + gz$ (internal energy, kinetic energy, potential energy, exactly as in our thermodynamics review, just written per unit mass now). The first law for a system is $dE_{sys}/dt = \dot{Q} - \dot{W}$, so the RTT gives the **energy equation** for a control volume:

$$\dot{Q} - \dot{W}_{shaft} - \dot{W}_{pressure} = \frac{\partial}{\partial t} \int_{CV} e \rho dV + \oint_{CS} \left(\hat{u} + \frac{V^2}{2} + gz \right) \rho (\mathbf{u} \cdot \mathbf{n}) dA,$$

where we have split the work into shaft work (done by a pump or turbine) and the work the surrounding pressure does in pushing fluid across the boundary, $\dot{W}_{pressure} = \oint p(\mathbf{u} \cdot \mathbf{n}) dA$. Moving this term to the flux side and using $h = \hat{u} + p/\rho$ (enthalpy, again from thermodynamics), for a steady flow through one inlet (1) and one outlet (2) this collapses to the familiar

$$\dot{Q} - \dot{W}_{shaft} = \dot{m} \left[h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1) \right],$$

which is precisely the energy balance we wrote down in Section 2, now derived rather than just quoted.

Bernoulli's equation as a special case. Suppose in addition the flow is steady, incompressible, inviscid (no friction, hence no losses) and there is no heat transfer and no shaft work. Then $\dot{Q} = \dot{W}_{shaft} = 0$. With no friction there is also no viscous dissipation, so the internal energy of the fluid does not change along a streamline, $\hat{u}_1 = \hat{u}_2$, and $h_2 - h_1$ reduces to $(p_2 - p_1)/\rho$. The energy equation becomes, after dividing by $\dot{m}$ and rearranging,

$$\frac{p_1}{\rho} + \frac{V_1^2}{2} + gz_1 = \frac{p_2}{\rho} + \frac{V_2^2}{2} + gz_2, \quad \text{or} \quad p + \frac{1}{2}\rho V^2 + \rho gz = \text{constant along a streamline.}$$

This is **Bernoulli's equation**. Notice everything it took to get here: steady, incompressible, inviscid, no shaft work, no heat transfer, evaluated along a single streamtube. It is really nothing but conservation of energy with friction, heat and work switched off, and it is worth remembering all of its assumptions, because every mistake students make with Bernoulli's equation comes from applying it where one of these assumptions fails (most commonly, across a pump, or in a flow with significant friction, or between two different streamlines in a rotational flow).

Physically, Bernoulli's equation says that along a streamline, pressure, kinetic energy and potential energy trade off against each other, but their sum is conserved. Where the streamlines squeeze together and the flow speeds up (recall $A_1 V_1 = A_2 V_2$ from continuity), the pressure must drop to pay for the increase in kinetic energy. This is precisely why the flow separates a shower curtain in towards you, why a curveball curves, and why an airplane wing generates lift: on the side where the streamlines are more crowded, the velocity is higher, and the pressure is lower.

8.5 Conservation of Angular Momentum

Many devices (pumps, turbines, sprinklers, mixers) exert their most important effect through a *torque*, not a force, so it is worth repeating the RTT argument one more time with $B = \mathbf{H}_O = \mathbf{r} \times m\mathbf{u}$, the angular momentum about a point O , so that $\beta = \mathbf{r} \times \mathbf{u}$. For a system, $d\mathbf{H}_{O,sys}/dt = \sum \mathbf{M}_O$, the net moment of all external forces about O . The RTT then gives the **angular momentum equation**,

$$\sum \mathbf{M}_O = \frac{\partial}{\partial t} \int_{CV} (\mathbf{r} \times \mathbf{u}) \rho dV + \oint_{CS} (\mathbf{r} \times \mathbf{u}) \rho (\mathbf{u} \cdot \mathbf{n}) dA,$$

which for steady flow with well-defined inlets and outlets reduces to the single scalar statement, about any fixed axis,

$$\sum M = \dot{m}_{out}(r_t V_t)_{out} - \dot{m}_{in}(r_t V_t)_{in},$$

where $r_t V_t$ is the moment of the tangential velocity about the axis at each port. This is exactly the same bookkeeping as the linear momentum equation, just with "moment of momentum" carried by the flow instead of momentum itself, and it is the equation we will use, unmodified, to analyze pumps and turbines in the turbomachinery chapter at the end of this handout: a rotating machine changes the angular momentum, not just the linear momentum, of the fluid passing through it, and the torque on the shaft is exactly the rate of that change.

Example: the lawn sprinkler. A simple sprinkler has two arms of length r , each ejecting water of density ρ at a flow rate $Q/2$ (so Q total) through a nozzle of area A at the tip, angled so the jet leaves tangent to the arm's rotation, in the plane of rotation. Take the CV to rotate with the sprinkler. Relative to the ground, if the sprinkler spins at ω , the absolute tangential velocity of the jet is $V_{rel} - r\omega$ (the jet velocity relative to the nozzle minus the speed the nozzle itself is carrying it backward, in the sense that opposes rotation), where $V_{rel} = (Q/2)/A$ from continuity in the rotating arm. With no external torque applied to the water other than at the central bearing, and nothing entering with angular momentum (the water enters along the axis, $r = 0$),

$$M_{shaft} = \dot{m}_{out} r V_{t,out} = \rho Q r \left(\frac{Q}{2A} - r\omega \right),$$

which is precisely the torque the bearing must resist to hold the sprinkler at speed ω ; setting $M_{shaft} = 0$ gives the free-spinning speed the sprinkler settles to if left completely unrestrained, exactly analogous to a rocket reaching a terminal spin rate.

9 Differential Analysis

The control volume equations of the last chapter are wonderful for engineering answers (forces, flow rates, power), but they throw away all the detail of what happens *inside* the control volume. If we want to know the velocity profile in a pipe, or the shape of a boundary layer, we need equations that hold at every point of the fluid. We get these by shrinking the control volume down to an infinitesimal box, exactly the way we did for the hydrostatic equation, except now the fluid is allowed to move.

9.1 A Review of Basic Integral Theorems from MIE230

Before we start, let us recall the two vector calculus results that make this chapter possible, both of which you saw in MIE230 (or an equivalent Calculus course).

The **divergence theorem** converts a surface integral of a vector field's flux into a volume integral of its divergence:

$$\oint_{CS} \mathbf{F} \cdot \mathbf{n} dA = \int_{CV} \nabla \cdot \mathbf{F} dV.$$

Physically, $\nabla \cdot \mathbf{F}$ measures the local rate at which $\mathbf{F}$ is "spreading out" of a point, and the theorem says that if you add up this spreading over every point inside a volume, you get exactly the net flow through the boundary, because whatever spreads out of one small cell flows straight into its neighbour, and all of those interior transfers cancel, leaving only the outer surface.

The second fact we need is simply that if a continuous function f satisfies $\int_V f dV = 0$ for *every* possible choice of volume V , no matter how small, then f must be zero everywhere. This sounds obvious, but it is the entire justification for turning integral balance laws into differential ones: shrink the control volume to a point.

9.2 Differential Continuity Equation

Apply the divergence theorem to the flux term of the integral continuity equation from the last chapter,

$$\int_{CV} \frac{\partial \rho}{\partial t} dV + \oint_{CS} \rho \mathbf{u} \cdot \mathbf{n} dA = \int_{CV} \frac{\partial \rho}{\partial t} dV + \int_{CV} \nabla \cdot (\rho \mathbf{u}) dV = \int_{CV} \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) \right] dV.$$

Since the control volume is completely arbitrary, and the left-hand side must equal zero for every choice of it, the integrand itself must vanish:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0.$$

This is the **differential continuity equation**. If we expand $\nabla \cdot (\rho \mathbf{u}) = \rho \nabla \cdot \mathbf{u} + \mathbf{u} \cdot \nabla \rho$ and recognize the material derivative from Section 5,

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0.$$

This form is very intuitive: $D\rho/Dt$ is the rate of change of density *following a fluid particle*, and it must be balanced by $\rho \nabla \cdot \mathbf{u}$, the rate at which the particle's volume is expanding (or contracting). If the particle's volume grows, its density must fall, and vice versa, exactly like squeezing or releasing a small balloon of fluid. For an **incompressible** flow the density of every particle stays fixed, $D\rho/Dt = 0$, so the continuity equation collapses to the beautifully simple

$$\nabla \cdot \mathbf{u} = 0, \quad \text{i.e.} \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0.$$

This says that the velocity field has no divergence: fluid does not pile up or thin out anywhere, whatever flows into any infinitesimal box must flow right back out.

9.3 The Stream Function

For a two-dimensional, incompressible flow, $\partial u/\partial x + \partial v/\partial y = 0$ from the last section, and this single constraint suggests a shortcut: instead of tracking two unknown functions $u(x, y)$ and $v(x, y)$ subject to one equation, why not describe the whole flow with a single scalar function that satisfies continuity automatically? Define the **stream function** $\psi(x, y)$ by

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}.$$

Substituting into continuity gives $\partial^2 \psi / \partial x \partial y - \partial^2 \psi / \partial y \partial x \equiv 0$, identically, for any smooth ψ at all. So any incompressible, two-dimensional velocity field can be represented this way, and any ψ we dream up automatically represents some valid incompressible flow, which is an enormous

convenience for building up solutions, exactly as we will do in the next chapter.

There is also a beautiful geometric meaning to ψ . Along a curve of constant ψ , $d\psi = \frac{\partial\psi}{\partial x}dx + \frac{\partial\psi}{\partial y}dy = -v dx + u dy = 0$, so $dy/dx = v/u$ along that curve, which is exactly the streamline equation from Section 5.3. *Lines of constant ψ are streamlines.* Furthermore, the volumetric flow rate (per unit depth) passing between two streamlines ψ_1 and ψ_2 is exactly $\psi_2 - \psi_1$, so the closer together the streamlines are drawn (for equal increments of ψ), the faster the local flow, giving an immediate, quantitative picture of speed from a streamline plot alone, without doing any further calculation.

9.4 Differential Momentum Equation

We repeat the argument of Section 8.3, but on an infinitesimal cube instead of a whole system, so that we can keep track of stresses that vary from face to face. In general a fluid element can feel a normal stress and *two* shear stresses on each of its six faces, and all of these together form the **stress tensor** σ_{ij} (the stress in direction j acting on the face whose normal points in direction i). Newton's second law applied to the mass $\rho dx dy dz$ of the element, using the same face-pair bookkeeping we used for pressure in the hydrostatics chapter (a net force per unit volume of $\partial\sigma_{ij}/\partial x_i$ from the stresses, in each direction j , plus the body force ρg_j), gives

$$\rho a_j = \rho \frac{Du_j}{Dt} = \frac{\partial\sigma_{ij}}{\partial x_i} + \rho g_j.$$

This is completely general, valid for any fluid, but it is not yet a usable equation because we have not related the stress to the motion of the fluid. For a fluid at rest, we already found in the hydrostatics chapter that $\sigma_{ij} = -p\delta_{ij}$ (pressure only, no shear). For a moving **Newtonian** fluid (recall from Section 4 that shear is proportional to the velocity gradient), the natural generalization to three dimensions is

$$\sigma_{ij} = -p\delta_{ij} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right),$$

which reduces to $\tau = \mu du/dy$ for the simple shear flow of Section 4, and to hydrostatics when $\mathbf{u} = 0$. Substituting this stress into Newton's second law, and using $\nabla \cdot \mathbf{u} = 0$ for an incompressible flow with constant μ (so that $\partial^2 u_i / \partial x_i \partial x_j$ terms vanish), we obtain the **Navier–Stokes equations**:

$$\rho \left[\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right] = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}.$$

Every term has a clear physical meaning, and it is worth reading the equation this way every single time you see it: *(mass) × (acceleration) equals (pressure force) plus (viscous force) plus (weight)*, per unit volume. The left side is just ρ times the acceleration we derived in Section 5, unsteady plus convective. On the right, $-\nabla p$ pushes fluid from high to low pressure (exactly as in hydrostatics), $\mu \nabla^2 \mathbf{u}$ is the net viscous force from neighbouring layers of fluid dragging on each other (a Laplacian measures how different a point is from the average of its neighbours, so this term smooths out velocity differences, exactly the way viscosity is supposed to), and $\rho \mathbf{g}$ is gravity.

Notice that the Navier–Stokes equations contain the nonlinear term $(\mathbf{u} \cdot \nabla) \mathbf{u}$ that we flagged back in Section 5. This term is why, apart from a short list of very simple geometries, nobody has ever solved these equations by hand, and why fields like CFD and turbulence modelling exist. If you drop the viscous term entirely ($\mu = 0$), you get the **Euler equation**, appropriate far from walls where viscosity has not had a chance to act, and Bernoulli’s equation can in fact be derived by integrating the Euler equation along a streamline, which is a more rigorous route to the same result we found by energy arguments in the last chapter.

9.5 Energy Equation

The same shrinking argument can be applied to the energy balance of Section 8.4. The result, for an incompressible flow with constant properties, is

$$\rho c_v \frac{DT}{Dt} = k \nabla^2 T + \Phi,$$

where k is the thermal conductivity (so $k \nabla^2 T$ is heat conduction, entering by the same neighbour-averaging logic as the viscous term above) and Φ is the **viscous dissipation**, a positive quantity built from the velocity gradients, representing the rate at which the friction we introduced in Section 4 irreversibly converts kinetic energy into heat. This term is exactly the mechanism that Bernoulli’s equation assumes away when it assumes $\hat{u}_1 = \hat{u}_2$; in a real (viscous) flow, dissipation always raises the internal energy at the expense of the mechanical energy, which is why every real flow has losses, and Bernoulli’s equation is always an idealization.

9.6 Vorticity and Irrotational Flow

Define the **vorticity** as the curl of the velocity field,

$$\boldsymbol{\omega} = \nabla \times \mathbf{u}.$$

To see what this measures, imagine placing a tiny paddle wheel (or a pair of perpendicular sticks) at a point in the flow and letting it be carried along and spun by the local velocity gradients. The vorticity is exactly twice the angular velocity of that little wheel. A flow is **irrotational** where $\boldsymbol{\omega} = 0$: a fluid particle there may translate and even deform, but it does not locally spin. Our stagnation-point flow example from Section 5 is in fact irrotational everywhere (you can check $\partial v/\partial x - \partial u/\partial y = 0$ for $u = kx, v = -ky$), while, by contrast, simple shear flow $u = cy, v = 0$ has $\omega_z = -c \neq 0$: even though the streamlines are straight, a paddle wheel placed in it would spin, because the fluid layer above pushes one side of the wheel harder than the layer below pushes the other side.

Vorticity is central to fluid mechanics for two reasons that will matter in the rest of the handout. First, it can be shown that in the absence of viscosity (and for a uniform-density flow starting from rest, or from a uniform stream, which is irrotational), the vorticity of a fluid particle never changes: vorticity is only *generated* by viscous effects, chiefly at solid walls. Second, where the flow is irrotational, $\nabla \times \mathbf{u} = 0$ implies that $\mathbf{u}$ can be written as the gradient of a scalar **velocity potential**, $\mathbf{u} = \nabla\phi$, and if the flow is also incompressible, substituting into $\nabla \cdot \mathbf{u} = 0$ gives Laplace's equation, $\nabla^2\phi = 0$. This is why, far from solid boundaries where viscosity has not yet had a chance to generate vorticity, we are often allowed to treat the flow as inviscid and irrotational (potential flow), which is a linear problem, dramatically simpler than the full Navier–Stokes equations. Right next to a solid wall, on the other hand, viscosity always matters, no matter how small μ is, and vorticity is created there; this is exactly the idea behind the boundary layer, which we will meet in the final chapter.

10 Potential Flow and Superposition

The last chapter closed with two remarkable facts about an inviscid, irrotational, incompressible flow: it can be described by a velocity potential ϕ with $\mathbf{u} = \nabla\phi$, and (in two dimensions) also by

a stream function ψ , and *both* of them satisfy Laplace's equation, $\nabla^2\phi = 0$ and $\nabla^2\psi = 0$. This is wonderful news, because Laplace's equation is **linear**: if ϕ_1 and ϕ_2 are each valid solutions, then so is any combination $\phi_1 + \phi_2$. Contrast this with the full Navier–Stokes equations, whose nonlinear convective term $(\mathbf{u} \cdot \nabla)\mathbf{u}$ has been the villain of this whole handout since Section 5, forbidding us from simply adding known solutions together to build new ones. In potential flow, that villain is gone (by assumption), and superposition becomes not just a convenience but the central technique of the entire chapter: build a library of a few simple flows, and add them together, in different amounts and positions, to approximate the flow around real, complicated shapes.

10.1 Elementary Plane Flows

Uniform stream. A uniform flow of speed U in the x direction has $u = U, v = 0$ everywhere, trivially satisfying $\nabla \times \mathbf{u} = 0$ and $\nabla \cdot \mathbf{u} = 0$. Integrating $u = \partial\phi/\partial x = U$ and $v = \partial\phi/\partial y = 0$ gives $\phi = Ux$, and similarly (using the definitions from Section 9.5) $\psi = Uy$.

Line source (or sink). Imagine fluid being injected radially outward, uniformly in all directions, from a line perpendicular to the page at the origin, at a volume flow rate per unit depth m (a **sink** is the same thing with $m < 0$, fluid draining inward). By symmetry the velocity is purely radial, $u_r = u_r(r)$, and continuity applied to a circle of radius r (the same mass must cross every circle, an idea straight out of Section 8.2) gives $u_r(2\pi r) = m$, so

$$u_r = \frac{m}{2\pi r}, \quad u_\theta = 0 \quad \implies \quad \phi = \frac{m}{2\pi} \ln r, \quad \psi = \frac{m}{2\pi} \theta,$$

where the potential follows from $\partial\phi/\partial r = u_r$ and the stream function from $\frac{1}{r}\partial\psi/\partial\theta = 0$, $-\partial\psi/\partial\theta = u_r$ appropriately adapted to polar form. Notice $u_r \rightarrow \infty$ as $r \rightarrow 0$: a real source is of course only an idealization of a finite injection region, valid everywhere except right at that one singular point.

Line (irrotational) vortex. Now imagine fluid moving in pure circles, $u_\theta(r)$, with $u_r = 0$, but demand that the flow be irrotational everywhere except at the origin, exactly the point-vortex idealization we need to reproduce the circulation Γ we already met in the airfoil theory of Section 13.7. Irrotationality in polar coordinates requires $\frac{1}{r}\frac{d}{dr}(ru_\theta) = 0$, so $ru_\theta = \text{const}$, and defining that constant through the circulation $\Gamma = \oint \mathbf{u} \cdot d\mathbf{s} = u_\theta(2\pi r)$,

$$u_\theta = \frac{\Gamma}{2\pi r}, \quad u_r = 0 \quad \implies \quad \phi = \frac{\Gamma}{2\pi} \theta, \quad \psi = -\frac{\Gamma}{2\pi} \ln r.$$

Notice that ϕ and ψ here are exactly swapped compared to the source, a nice illustration of the fact that ϕ and ψ always play complementary roles: the streamlines of a source (radial rays) are the equipotentials of a vortex, and vice versa. Also notice that u_θ blows up at the center exactly like a source's u_r does, again only a mathematical idealization; a real vortex (a tornado, the flow leaving your bathtub drain) has a small rotational core near the center where viscosity intervenes and u_θ instead grows linearly with r , like a rigid body, before matching onto this irrotational, decaying outer solution.

10.2 Superposition: the Rankine Half-Body

As our first taste of superposition, add a uniform stream U (flowing in the $+x$ direction) to a source of strength m placed at the origin:

$$\psi = Uy + \frac{m}{2\pi}\theta = Ur \sin \theta + \frac{m}{2\pi}\theta.$$

Far upstream, the uniform stream dominates and the flow looks undisturbed. Near the origin, the source dominates and pushes fluid radially outward. Somewhere in between there must be a **stagnation point**, where the source's outward push exactly cancels the stream's forward push; setting $u_r = U \cos \theta + m/(2\pi r) = 0$ (only the radial component of the uniform stream contributes to u_r) along $\theta = \pi$ (directly upstream) gives the stagnation point at $r = a = m/(2\pi U)$. The dividing streamline that passes through this stagnation point (found by setting ψ equal to its value there, $\psi = m/2$ on $\theta = \pi$) separates the source fluid, which is trapped forever inside a bulging, open-ended contour, from the stream fluid, which flows around the outside. This contour, called the **Rankine half-body**, is a legitimate, closed (on one end) shape, and since the outside of it is a perfectly valid potential flow solution, we have just found, almost for free, the inviscid flow around a rounded, bullet-like nose, of exactly the sort you would want for a streamlined strut or a pitot-tube casing.

10.3 Flow Past a Circular Cylinder

Now superpose a uniform stream with a **doublet**: the limiting case of a source-sink pair as their separation shrinks to zero while the product of their strength and spacing stays fixed at κ (a doublet by itself represents, for instance, the way a source's fluid must go somewhere and is immediately

reabsorbed close by). Its stream function is $\psi_{doublet} = -\kappa \sin \theta / r$. Adding a uniform stream,

$$\psi = Ur \sin \theta - \frac{\kappa}{r} \sin \theta = \left(Ur - \frac{\kappa}{r} \right) \sin \theta.$$

Setting $\psi = 0$ gives either $\sin \theta = 0$ (the x -axis) or $r = \sqrt{\kappa/U} \equiv R$, a full circle. Since $\psi = 0$ is itself a valid streamline (it must be, being a solution of the same linear equation), and a streamline can never be crossed (Section 5.3), we have just constructed, exactly, the potential flow around a solid circular cylinder of radius R , simply by choosing $\kappa = UR^2$:

$$\psi = U \left(r - \frac{R^2}{r} \right) \sin \theta, \quad u_r = U \left(1 - \frac{R^2}{r^2} \right) \cos \theta, \quad u_\theta = -U \left(1 + \frac{R^2}{r^2} \right) \sin \theta.$$

On the cylinder surface $r = R$, $u_r = 0$ (as it must, no flow through a solid wall) and the surface speed is purely tangential, $u_\theta|_R = -2U \sin \theta$, reaching a maximum speed of $2U$ at the top and bottom ($\theta = \pm\pi/2$) and zero at the front and back stagnation points ($\theta = 0, \pi$). Applying Bernoulli's equation (Section 8.4, valid here since the whole flow is inviscid, incompressible and irrotational) between the far upstream condition and the surface,

$$p_s = p_\infty + \frac{1}{2} \rho U^2 [1 - 4 \sin^2 \theta], \quad C_p \equiv \frac{p_s - p_\infty}{\frac{1}{2} \rho U^2} = 1 - 4 \sin^2 \theta.$$

This pressure distribution is symmetric both fore-aft and top-bottom, and if you integrate it around the entire cylinder to find the net force, every contribution cancels and you get *exactly zero* drag. This famous, and famously wrong, result is **D'Alembert's paradox**: potential theory insists a body can move through an inviscid fluid forever without any resistance at all, in flagrant contradiction with everyday experience. The resolution is exactly the boundary layer story of Section 13.6: real flow separates on the rear half of the cylinder, where potential theory demands a strong adverse pressure gradient (pressure recovering from its low point at the shoulders back up to the front-stagnation value at the rear), and separation destroys the rear-facing pressure recovery that the paradox depends on, leaving a low-pressure wake and a large pressure drag. D'Alembert's paradox is not a mistake in the mathematics, it is a sharp, precise statement of exactly how much physics gets thrown away the moment we set $\mu = 0$.

10.4 Cylinder with Circulation and the Magnus Effect

Finally, superpose a line vortex of strength Γ (centered at the origin, so it does not disturb the $\psi = 0$ circle we just built) onto the cylinder flow above:

$$\psi = U \left(r - \frac{R^2}{r} \right) \sin \theta - \frac{\Gamma}{2\pi} \ln r.$$

Adding a vortex centered exactly at the cylinder's center does not change the shape of the $r = R$ streamline (a circle remains a valid streamline for any strength of a concentric vortex, by symmetry) so this is still a legitimate flow around the same solid cylinder, but now with an extra swirl added on top. The surface velocity becomes $u_\theta|_R = -2U \sin \theta - \Gamma/(2\pi R)$, and Bernoulli's equation now gives an asymmetric pressure distribution, higher on the side where the vortex opposes the stream's tangential velocity, lower on the side where it reinforces it. Integrating this pressure around the cylinder (a short calculation using $F_y = -\oint p \sin \theta R d\theta$, keeping only the cross term that survives the angular integration) gives, remarkably, a clean result with no arbitrary constants at all:

$$L' = \rho U \Gamma,$$

exactly the **Kutta–Joukowski** lift formula we already quoted (without derivation) back in Section 13.7, now derived rigorously for a cylinder, and it turns out to hold for a body of *any* shape carrying circulation Γ in a stream U . Physically, adding circulation speeds up the flow on one side of the body and slows it on the other (recall the sketch of crowded streamlines in Section 8.4), the higher speed side has lower pressure by Bernoulli, and the pressure difference integrates to a net force perpendicular to the stream. This is the **Magnus effect**, responsible for the curve on a spinning baseball or golf ball (where the spinning surface drags a thin layer of real, viscous fluid around with it, generating an effective circulation) and for the entire principle by which an airfoil, through the sharp trailing edge enforcing the Kutta condition, selects exactly the amount of circulation needed to produce the lift we calculated from thin airfoil theory.

11 Dimensional Analysis

11.1 Essence of Dimensional Analysis

Look back at the Navier–Stokes equations from the last chapter. They are, in general, impossible to solve by hand. And yet, an engineer facing a new problem, say the drag on a new car shape, cannot simply give up. Dimensional analysis is the tool that lets us make progress *without solving the governing equations at all*, just by insisting that any physical relationship must be dimensionally consistent.

Here is the core idea. Suppose the drag force F on a body depends on its size L , the fluid speed V , the density ρ and the viscosity μ : $F = f(L, V, \rho, \mu)$. This is a relationship between five dimensional quantities. But every equation of physics must balance dimensionally (you cannot add kilograms to meters), and it turns out that this requirement is so restrictive that it forces the relationship to collapse into a relationship between a much smaller number of *dimensionless* groups. The immediate practical payoff is enormous: instead of running experiments varying F against four independent variables separately (an enormous, expensive test matrix), we only need to find how one dimensionless group depends on another, a single curve. This is also exactly the logic that makes wind tunnel testing and scale models possible, which is why the next chapter opens with wind tunnels: if we match the right dimensionless groups between a small model and a full-size prototype, the physics itself is guaranteed to match.

11.2 The Buckingham Π Theorem

The formal statement of this idea is the **Buckingham Π theorem**: if a physical relationship involves n dimensional variables, and these variables involve k independent fundamental dimensions (in fluid mechanics, usually mass M , length L and time T), then the relationship can be rewritten in terms of exactly $n - k$ independent dimensionless groups, conventionally called Π groups.

The method (the “repeating variables” method) works as follows. List all n variables and their dimensions. Choose k of them as **repeating variables**: they should together contain all k fundamental dimensions, and should not by themselves be combinable into a dimensionless group. Then form each Π group by combining the repeating variables, raised to unknown exponents, with one of the remaining $n - k$ variables, and solve for the exponents that make the whole combination dimensionless.

Example: drag on a sphere. Take $F = f(L, V, \rho, \mu)$, so $n = 5$ variables, with dimensions $[F] = MLT^{-2}$, $[L] = L$, $[V] = LT^{-1}$, $[\rho] = ML^{-3}$, $[\mu] = ML^{-1}T^{-1}$. There are $k = 3$ fundamental dimensions (M, L, T), so we expect $n - k = 2$ dimensionless groups. Choose L, V, ρ as repeating variables (together they contain M, L, T , and none of them alone is dimensionless). Form Π_1 from F :

$$\Pi_1 = FL^a V^b \rho^c, \quad [\Pi_1] = (MLT^{-2})(L^a)(LT^{-1})^b(ML^{-3})^c = M^{1+c} L^{1+a+b-3c} T^{-2-b}.$$

Setting each exponent to zero: from T , $b = -2$; from M , $c = -1$; from L , $1+a-2+3 = 0 \Rightarrow a = -2$. So

$$\Pi_1 = \frac{F}{\rho V^2 L^2}.$$

This is nothing but a force nondimensionalized by a "dynamic pressure times area" scale, and it is essentially the definition of a **drag coefficient**, $C_D = F/(\frac{1}{2}\rho V^2 A)$, up to the constant $\frac{1}{2}$ and the exact power of L used for area, which are conventions. Repeating the same procedure with μ instead of F gives (you should check this yourself as practice)

$$\Pi_2 = \frac{\rho V L}{\mu} = \text{Re},$$

which you will recognize as the **Reynolds number**. The Buckingham theorem then guarantees that the original five-variable relationship can be written as a relationship between only two dimensionless numbers,

$$\frac{F}{\rho V^2 L^2} = g(\text{Re}),$$

for some function g that dimensional analysis cannot tell us (that is what experiments, or CFD, are for), but whose *existence and form* we now know for free. This is the entire reason drag coefficients are always plotted against Reynolds number, and never against F, V, ρ and μ separately.

11.3 Non-dimensional Flow Equations

The same idea can be applied directly to the Navier–Stokes equation itself, and this is arguably more illuminating than the Buckingham method because it tells us exactly *which* physical effects each dimensionless group compares. Introduce a characteristic length L , velocity U , and time L/U ,

and define nondimensional variables (starred) by $\mathbf{x}^* = \mathbf{x}/L$, $\mathbf{u}^* = \mathbf{u}/U$, $t^* = tU/L$, $p^* = p/(\rho U^2)$. Substituting into

$$\rho \left[\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right] = -\nabla p + \mu \nabla^2 \mathbf{u}$$

and carrying through the chain rule (each derivative $\partial/\partial x \rightarrow (1/L)\partial/\partial x^*$, and so on) gives, after dividing everything by $\rho U^2/L$,

$$\frac{\partial \mathbf{u}^*}{\partial t^*} + (\mathbf{u}^* \cdot \nabla^*) \mathbf{u}^* = -\nabla^* p^* + \frac{1}{\text{Re}} \nabla^{*2} \mathbf{u}^*, \quad \text{Re} = \frac{\rho U L}{\mu}.$$

Every term in this equation is now of order one, by construction, so the *only* place the physical parameters ρ, U, L, μ can enter is through the single number Re multiplying the viscous term. This tells us immediately, and rigorously, that two flows around geometrically similar bodies, with different fluids and different sizes and speeds, will have identical (dimensionless) velocity and pressure fields provided they share the same Re . This is called **dynamic similarity**, and it is exactly the principle that lets a small wind tunnel model, tested at a matched Reynolds number, predict the full-scale behaviour of, say, an airplane.

11.4 Some Numbers To Remember & Flow Similarity

A handful of dimensionless groups recur constantly in this course and beyond, and it is worth having physical pictures for each of them, in the same spirit as we built one for the Reynolds number above.

- **Reynolds number**, $\text{Re} = \rho U L / \mu = U L / \nu$ (where $\nu = \mu / \rho$ is the kinematic viscosity), compares inertial forces to viscous forces. Look again at the nondimensional Navier–Stokes equation: the convective term is order one, while the viscous term is scaled by $1/\text{Re}$. At low Re , viscosity dominates everywhere and the flow is smooth and orderly (creeping flow); at high Re , inertia dominates almost everywhere, viscosity matters only in thin layers near walls, and the flow is prone to instability and turbulence.
- **Froude number**, $Fr = U / \sqrt{gL}$, compares inertial forces to gravity, and governs flows with a free surface, such as waves and open channel flow, where the shape of the free surface depends on gravity.
- **Mach number**, $M = U / c$ where c is the speed of sound, compares the flow speed to how fast pressure disturbances can propagate, and governs compressibility: recall from the hydrostatics

chapter that we could integrate $dp/dz = -\rho g$ directly only because we assumed constant density; the same idea applies in a moving flow, and it turns out density variations become important once M exceeds about 0.3.

- **Weber number**, $We = \rho U^2 L / \sigma$ (with σ the surface tension), compares inertia to surface tension, and matters for droplets, bubbles and thin films.

For two flows to be fully **dynamically similar**, they need matching values of every dimensionless group that is relevant to the problem, in addition to being **geometrically similar** (same shape, all lengths scaled by the same factor) and, for unsteady flows, **kinematically similar** (streamline patterns that are simply scaled versions of each other). In practice it is often impossible to match every group at once with a small model (matching Re with a smaller model demands a much higher velocity, which might push M too high), and part of the art of experimental fluid mechanics is deciding which similarity parameter matters most for the physics you are trying to capture, and living with a mismatch in the others.

12 Internal Flows

12.1 Understanding Flow Regimes

Osborne Reynolds' famous dye experiment showed that the same fluid, in the same pipe, can behave in two completely different ways depending on speed. At low speed, a thin streak of dye injected into the pipe stays as a thin, straight line: this is **laminar** flow, where the fluid genuinely moves in smooth layers (recall the box-stack picture from Section 4), and momentum is transferred between layers purely by molecular viscosity. Beyond some critical speed, the dye streak suddenly breaks up and mixes violently throughout the cross section: this is **turbulent** flow, a chaotic, three-dimensional, unsteady state full of eddies of many different sizes, in which momentum is transferred mainly by the physical churning of fluid parcels, a far more effective mixing mechanism than pure viscosity.

Whether a given flow is laminar or turbulent is controlled by the Reynolds number we built in the last chapter, $Re = \rho V D / \mu$ for a pipe of diameter D , and this is exactly why the Reynolds number kept reappearing: it literally measures the ratio of inertial to viscous forces, and instabilities that would otherwise be damped out by viscosity survive and grow once inertia dominates enough. For

flow in a pipe, experiments show that the flow is essentially always laminar below $Re \approx 2300$, and (for the practical case of an ordinary pipe with some roughness and disturbances) is fully turbulent above roughly $Re \approx 4000$, with a transitional range in between.

12.2 Internal Viscous Flow: Entrance Region and Fully Developed Flow

When a fluid enters a pipe from a reservoir, the velocity profile is at first nearly uniform. As the fluid moves downstream, the no-slip condition at the wall (the fluid right at a solid surface must move with the surface, one of the most important boundary conditions in this course) forces the fluid there to slow to zero, and this retardation gradually spreads inward due to viscosity, exactly as the viscous term $\mu \nabla^2 \mathbf{u}$ diffuses momentum in the Navier–Stokes equation. This developing region is called the **entrance length**, L_e , after which the velocity profile stops changing with x and the flow is said to be **fully developed**. For laminar flow, $L_e/D \approx 0.06 Re$; for turbulent flow the mixing is much more effective and L_e/D is typically only around 10 to 60, essentially independent of Re .

12.3 Some Analytical Solutions for Laminar Developed Pipe Flow

Fully developed flow is one of the few cases where we can actually solve the Navier–Stokes equations by hand, precisely because the assumptions kill the troublesome nonlinear term. Consider steady, incompressible, laminar flow in a straight horizontal pipe of radius R . By symmetry the flow only has an axial velocity component $u(r)$ (no radial or swirl velocity), and "fully developed" means $\partial u / \partial x = 0$. Look back at the acceleration from Section 5: with $v = w = 0$ and $\partial u / \partial x = 0$ and steady flow, every single term in $D\mathbf{u}/Dt$ vanishes identically. The fluid is not accelerating at all, even though it is clearly moving; the pressure force pushing it forward is in perfect local balance with the viscous drag holding it back. The axial component of the Navier–Stokes equation (in cylindrical coordinates, with $\nabla^2 u = \frac{1}{r} \frac{d}{dr} \left(r \frac{du}{dr} \right)$ for this flow) then reduces to

$$0 = -\frac{\partial p}{\partial x} + \frac{\mu}{r} \frac{d}{dr} \left(r \frac{du}{dr} \right).$$

Since u depends only on r while p depends only on x , the only way this equation can hold for all r and x is if both sides are equal to the same constant, $dp/dx = \text{const}$: the pressure must fall linearly along the pipe, which makes complete physical sense, since a constant pressure gradient is exactly

what is needed to push against a constant viscous retarding force. Integrating twice,

$$\frac{d}{dr} \left(r \frac{du}{dr} \right) = \frac{1}{\mu} \frac{\partial p}{\partial x} r \implies u(r) = \frac{1}{4\mu} \frac{\partial p}{\partial x} r^2 + C_1 \ln r + C_2.$$

Physically, the velocity must stay finite at the centerline $r = 0$, which forces $C_1 = 0$ (else $\ln r \rightarrow -\infty$), and the no-slip condition $u(R) = 0$ at the wall gives $C_2 = -\frac{1}{4\mu} \frac{dp}{dx} R^2$. Hence

$$u(r) = -\frac{1}{4\mu} \frac{\partial p}{\partial x} (R^2 - r^2),$$

a parabola, known as **Poiseuille flow**, with the maximum velocity $u_{max} = -\frac{1}{4\mu} \frac{dp}{dx} R^2$ at the center. The (negative) sign of dp/dx is exactly what makes u positive: pressure must decrease in the direction of flow.

Integrating this profile over the cross-section gives the volumetric flow rate,

$$Q = \int_0^R u(r) 2\pi r dr = -\frac{\pi R^4}{8\mu} \frac{\partial p}{\partial x} = \frac{\pi R^4 \Delta p}{8\mu L},$$

where Δp is the pressure drop over a length L . This is the **Hagen–Poiseuille law**, and notice the very strong R^4 dependence: halving the radius of a pipe cuts the flow rate (for a fixed pressure drop) by a factor of sixteen, which is exactly why, physiologically, small changes in blood vessel diameter have such an outsized effect on blood flow. Dividing Q by the area πR^2 gives the mean velocity $V = u_{max}/2$: in laminar pipe flow, the average speed is exactly half the centerline speed, a clean fact that follows purely from the parabola having a specific, uniform curvature.

12.4 Head Loss

For turbulent flow, or for more complicated geometries, we cannot integrate the Navier–Stokes equations by hand, and we fall back on the energy equation of Section 8.4. Write Bernoulli's equation between two points in a pipe, but now add a term h_L (the **head loss**, energy per unit weight lost to friction) to honestly account for the fact that real flows are viscous and Bernoulli's equation, as we stressed in Section 8.4, assumed that away:

$$\frac{p_1}{\rho g} + \frac{V_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 + h_L.$$

This h_L is exactly the viscous dissipation Φ from Section 9.4, integrated along the pipe and expressed as an equivalent drop in mechanical energy. For flow in a straight pipe of length L and diameter D , dimensional analysis (following exactly the Buckingham method of the last chapter, now applied to $\Delta p = f(L, D, V, \rho, \mu, \varepsilon)$ with ε the wall roughness) shows that the pressure drop must take the form

$$\Delta p = f \frac{L}{D} \frac{\rho V^2}{2}, \quad h_L = \frac{\Delta p}{\rho g} = f \frac{L}{D} \frac{V^2}{2g},$$

which is the **Darcy–Weisbach equation**, where f , the **Darcy friction factor**, is the one remaining dimensionless group and depends on Re and the relative roughness ε/D .

For laminar flow we do not even need to guess f : comparing the Darcy–Weisbach equation to the Hagen–Poiseuille law derived above (rewritten using $Q = VA = V\pi D^2/4$) gives, after a short algebra check you should do yourself,

$$f = \frac{64}{\text{Re}} \quad (\text{laminar}),$$

independent of roughness, since in smooth, layered flow the wall’s microscopic bumps are simply engulfed within the slow-moving fluid right next to the wall and never get ”felt” by the bulk flow.

12.5 Turbulence Modelling

For turbulent flow, the velocity at any point is not steady even in a ”steady” flow, it fluctuates chaotically about a mean value. Following the idea of Reynolds decomposition, we split $u = \bar{u} + u'$ (mean plus fluctuation, with $\overline{u'} = 0$ by definition) and substitute into the Navier–Stokes equations, then average. Because the equations are nonlinear (again, that same convective term from Section 5), the average of the equation for $\bar{u}$ picks up an extra term built from the fluctuations, $-\rho \overline{u'v'}$, called the **Reynolds stress**. Physically, this represents the extra transfer of momentum caused by turbulent eddies physically carrying fluid (and the momentum it carries) from one layer to another, on top of the transfer already accounted for by molecular viscosity μ .

The trouble is that this averaging procedure introduces new unknowns (the Reynolds stresses) without introducing new equations for them; this is the famous **closure problem** of turbulence. Engineers close the system with a turbulence model, the simplest and most influential idea being the **eddy viscosity** concept: by analogy with $\tau = \mu du/dy$, we model $-\rho \overline{u'v'} = \mu_t d\bar{u}/dy$, where μ_t

is not a fluid property but a property of the flow itself (it depends on how vigorous the turbulence is, and is typically many times larger than μ), which is what models like k - ε or k - ω are ultimately trying to estimate. The net practical effect of turbulence, for this course, is simply that momentum gets mixed across the pipe far more effectively than in laminar flow, giving a much flatter (fuller) velocity profile than the laminar parabola, with a thin region near the wall where the steep gradients (and hence the wall shear stress) are concentrated.

12.6 Darcy Friction Factor and The Moody Chart

For turbulent flow, f cannot be derived analytically the way we derived $f = 64/\text{Re}$ for laminar flow, because the eddy viscosity above is not a known fluid property, and f is instead obtained from experimental correlations, most famously the implicit Colebrook equation,

$$\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\varepsilon/D}{3.7} + \frac{2.51}{\text{Re}\sqrt{f}} \right),$$

whose results are plotted for convenience on the **Moody chart**, giving f as a function of Re for a family of curves, one per relative roughness ε/D . Reading the chart from left (laminar, where all curves collapse onto the single line $f = 64/\text{Re}$, exactly matching our analytical result and confirming that roughness truly does not matter there) to right (fully rough turbulent flow, where the curves flatten out and f becomes independent of Re , because at high enough speed the roughness elements themselves, rather than viscosity, are what generates the dominant momentum losses) tells the whole physical story of pipe friction in one picture. Figure 5 shows the chart, computed by solving the Colebrook equation numerically at each Re for a family of fixed ε/D values, exactly as you would if you did not want to read values off a printed chart by eye.

12.7 Minor Losses

Any pipe fitting that is not a straight, constant-diameter run (a valve, an elbow, a sudden expansion, an entrance or exit) causes additional dissipation beyond the wall friction we have just discussed, because it forces the flow to accelerate, decelerate, separate, or otherwise depart from the simple fully developed profile of Section 12.3. These are called **minor losses** (a slightly misleading name, since in a short pipe system they can easily dominate the total loss), and by exactly the same

dimensional reasoning that gave us the Darcy–Weisbach equation, each is modelled as

$$h_{L,minor} = K_L \frac{V^2}{2g},$$

where K_L is an empirically determined **loss coefficient** specific to that fitting. One special case we can actually derive rather than measure is the **sudden expansion**, where applying the momentum equation of Section 8.3 (using the pressure p_1 acting on the whole larger area A_2 just downstream of the step, since the flow separates and the low-speed corner region is essentially at the upstream pressure) together with the energy equation gives the exact result $K_L = (1 - A_1/A_2)^2$, a rare case where a minor loss coefficient follows directly from first principles rather than from a curve fit.

12.8 Pipe Networks

Finally, real piping systems consist of many pipes in series and in parallel, and analyzing them is nothing more than applying continuity (Section 8.2) and the energy equation with losses (Section 12.4) systematically. For pipes in **series**, the same flow rate Q passes through every pipe (continuity, since none of the fluid has anywhere else to go), and the total head loss is the sum of the head losses of every pipe and fitting, $h_{L,total} = \sum h_{L,i}$, exactly analogous to resistors in series (each $h_L \propto Q^2$ takes the place of IR , with Q playing the role of current). For pipes in **parallel** between two common junctions, every pipe must have the *same* head loss between those two points, since the pressure (and elevation) at each junction is a single shared value, so $h_{L,1} = h_{L,2} = \dots$, while the total flow rate splits between the branches, $Q_{total} = \sum Q_i$ (continuity again). This is exactly analogous to resistors in parallel, and the same systematic bookkeeping, continuity at every junction plus a consistent head loss around every loop, extends directly to arbitrarily large networks, forming the basis of the methods used to design municipal water distribution systems.

13 External Flows

13.1 Essence of Wind Tunnel Testing & CFD

When a body sits in an unbounded stream, rather than being confined by pipe walls, we call it an external flow, and the central engineering questions become the lift and drag it experiences. As we saw in Section 10, we cannot solve the Navier–Stokes equations by hand for a shape as complicated

as a wing or a car, and the two tools engineers use instead are wind tunnel testing and computational fluid dynamics (CFD).

A wind tunnel works because of dynamic similarity (Section 10.4): if a small-scale model is geometrically similar to the full-size body, and is tested at the same Reynolds number (and Mach number, if compressibility matters), then the nondimensional flow field around the model, in particular the drag coefficient C_D and lift coefficient C_L , will be identical to the full-scale case, even though the physical size, speed, and even the fluid may all differ. This is an enormous practical convenience, but as we noted, matching every dimensionless group simultaneously is often impossible with a small model, and this is one of the main sources of uncertainty in wind tunnel testing.

CFD instead solves a discretized (finite volume, finite element, or finite difference) approximation of the Navier–Stokes equations directly on a computer, over a mesh of small cells that plays the same conceptual role as the infinitesimal control volumes we shrank down to in Section 9. In fully turbulent flow, since we cannot resolve every eddy down to the smallest scales without an enormous mesh, CFD almost always relies on some version of the turbulence modelling idea from Section 12.6 (an eddy viscosity, or a more sophisticated closure) to represent the effect of the unresolved turbulent fluctuations. Both experiment and CFD are, in the end, complementary ways of approximating the same underlying physics that this whole handout has been building towards.

13.2 Boundary Layer Theory

Consider a uniform stream U_∞ flowing over a flat plate. Far from the plate, the flow behaves as if it were completely inviscid (recall the potential flow idea from Section 9.5), since the velocity gradients are small there. But right at the plate’s surface, the no-slip condition (Section 12.2) forces the velocity to be exactly zero, no matter how small μ is. So there must be some thin region near the wall where the velocity rises from 0 to (nearly) U_∞ , and it is inside this thin region, called the **boundary layer**, that essentially all of the viscous action, and all of the vorticity generation we mentioned in Section 9.5, is concentrated.

Deriving the boundary layer equations. Start from the full, steady, two-dimensional Navier–Stokes and continuity equations (Section 9.3) with no pressure gradient (a flat plate) and no gravity:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

Inside a thin layer of thickness $\delta \ll x$, the velocity u changes by an amount $\sim U_\infty$ over the short distance δ in y , but only over the much longer distance x in the streamwise direction. So $\partial u/\partial y \sim U_\infty/\delta$ is much larger than $\partial u/\partial x \sim U_\infty/x$, and comparing the two diffusive terms, $\partial^2 u/\partial y^2 \sim U_\infty/\delta^2$ completely dominates $\partial^2 u/\partial x^2 \sim U_\infty/x^2$ since $\delta \ll x$. Dropping the negligible streamwise diffusion term, and a matching order-of-magnitude argument on the y -momentum equation (which shows $\partial p/\partial y \approx 0$ across such a thin layer, i.e. the pressure is imposed on the layer by the outer, inviscid flow and does not vary through its thickness) gives **Prandtl's boundary layer equations**:

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{dp}{dx} + \nu \frac{\partial^2 u}{\partial y^2}, \quad \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0,$$

a considerable simplification of the full Navier–Stokes equations (a parabolic, rather than elliptic, system, meaning information only propagates downstream, exactly consistent with the picture of a boundary layer developing progressively along a surface).

13.3 The Blasius Solution

For a flat plate, $dp/dx = 0$ (the outer flow stays at the uniform speed U_∞), and the equations become

$$u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = \nu \frac{\partial^2 u}{\partial y^2}.$$

Blasius noticed that this problem has no natural length scale of its own (a semi-infinite flat plate looks the same at any magnification, apart from the distance x already measured along it), which suggests the velocity profile at every station x is really the *same* shape, merely stretched in y by the local boundary layer thickness. This motivates a **similarity variable** that combines y and x into one:

$$\eta = y \sqrt{\frac{U_\infty}{\nu x}},$$

exactly the ratio $y/\delta(x)$ suggested by our earlier estimate $\delta \sim \sqrt{\nu x/U_\infty}$ in the last section. Introduce the stream function (Section 9.2, since this flow is two-dimensional and incompressible) in the self-similar form $\psi = \sqrt{\nu U_\infty x} f(\eta)$, chosen so that

$$u = \frac{\partial \psi}{\partial y} = U_\infty f'(\eta),$$

i.e. $f'(\eta)$ is simply the velocity profile shape, normalized to run from 0 at the wall to 1 far outside the layer. Computing $v = -\partial\psi/\partial x$ using the chain rule (differentiating $\psi(x, \eta(x, y))$, remembering that η itself depends on x) gives $v = \frac{1}{2}\sqrt{\nu U_\infty/x}(\eta f' - f)$. Substituting u and v into the boundary layer momentum equation above, and converting every x - and y -derivative into η -derivatives via the chain rule (a somewhat lengthy but entirely mechanical exercise in differentiation, which we simply carry out term by term), every explicit dependence on x cancels out completely, exactly confirming the similarity hypothesis, and leaves a single ordinary differential equation, the **Blasius equation**:

$$2f''' + ff'' = 0,$$

with boundary conditions $f(0) = 0$ (no flow through the wall, $v = 0$ at $y = 0$), $f'(0) = 0$ (no-slip, $u = 0$ at $y = 0$), and $f'(\infty) = 1$ (matching the outer stream, $u \rightarrow U_\infty$). This equation is nonlinear (that same convective-term nonlinearity from Section 5, which cannot be escaped just by simplifying to a boundary layer) and it has no known closed-form solution; Blasius solved it by numerical shooting (guessing $f''(0)$, integrating outward, and adjusting the guess until $f' \rightarrow 1$ correctly), the standard technique used to this day for exactly this kind of two-point boundary value problem. The numerical solution gives, to four figures,

$$f''(0) = 0.332,$$

and $f'(\eta)$ reaches 0.99 at $\eta \approx 5.0$, precisely the origin of the constant $\delta \approx 5.0x/\sqrt{\text{Re}_x}$ we quoted without derivation in the last section: now we can see exactly where that number comes from, a numerically integrated nonlinear ODE, not a guess.

Since $\tau_w = \mu(\partial u/\partial y)|_{y=0}$ and $\partial u/\partial y = U_\infty f''(\eta)\sqrt{U_\infty/(\nu x)}$ by the chain rule, evaluating at $\eta = 0$ gives

$$\tau_w = \mu U_\infty \sqrt{\frac{U_\infty}{\nu x}} f''(0) = 0.332 \rho U_\infty^2 \sqrt{\frac{\nu}{U_\infty x}} \quad \Rightarrow \quad c_f \equiv \frac{\tau_w}{\frac{1}{2}\rho U_\infty^2} = \frac{0.664}{\sqrt{\text{Re}_x}},$$

the exact result we used, again without derivation, in Section 13.5. Integrating the profile $f'(\eta)$ numerically over η also gives the exact displacement and momentum thicknesses (defined properly in the next section), $\delta^* = 1.721\sqrt{\nu x/U_\infty}$ and $\theta = 0.664\sqrt{\nu x/U_\infty}$, so that the ratio $H = \delta^*/\theta = 2.59$ for the Blasius profile, a number we will use as a benchmark in a moment.

13.4 The Momentum Integral Equation, Derived

The Blasius solution is exact but only exists (even numerically) for the simplest case of zero pressure gradient. The moment a real body's curvature makes $dp/dx \neq 0$, we generally cannot solve the boundary layer PDE at all, exactly as with the full Navier–Stokes equations, and we fall back on the control-volume idea of Section 8 to get an approximate, but far more broadly applicable, ordinary differential equation instead.

Take a control volume of fixed height h (chosen large enough that $h > \delta$ at every x under consideration, so the top face sits entirely in the outer, inviscid flow) and thickness dx , exactly as sketched in Figure 6. Since h is fixed (independent of x), we may differentiate integrals over $y \in [0, h]$ directly, with no extra boundary terms to track. Mass entering the left face is $\int_0^h \rho u \, dy$, and by continuity, whatever extra mass leaves through the right face over what entered on the left must have entered through the top, at the (uniform, outer-flow) velocity U :

$$\dot{m}_{top} = \frac{d}{dx} \left(\int_0^h \rho u \, dy \right) dx,$$

which carries into the control volume an x -momentum flux of $U \dot{m}_{top}$. Newton's second law (Section 8.3) in the x direction, with the only forces being pressure on the two vertical faces and wall shear τ_w on the bottom (the top face, sitting in inviscid flow, carries no shear), reads

$$\frac{d}{dx} \left(\int_0^h \rho u^2 \, dy \right) dx - U \frac{d}{dx} \left(\int_0^h \rho u \, dy \right) dx = -h \frac{dp}{dx} dx - \tau_w dx.$$

The outer, inviscid flow obeys Bernoulli's equation (Section 8.4) along its own streamline just above the layer, so $dp/dx = -\rho U \, dU/dx$ (the compressible-free Euler equation of Section 9.3), giving $-h \, dp/dx = h \rho U \, dU/dx$ on the right.

Now introduce the two thicknesses that will let us package these integrals compactly:

$$\delta^*(x) = \int_0^h \left(1 - \frac{u}{U} \right) dy, \quad \theta(x) = \int_0^h \frac{u}{U} \left(1 - \frac{u}{U} \right) dy,$$

(the upper limit may equally be written ∞ or δ , since $u = U$ identically above the layer and the integrands vanish there). The displacement thickness δ^* is, physically, exactly how far the outer streamlines are pushed outward because the layer carries a mass-flow deficit $\int \rho(U - u) \, dy$ compared

to a uniform stream of speed U over the same height; the momentum thickness θ is the analogous deficit in momentum flux, $\int \rho u(U - u)dy = \rho U^2 \theta$.

With $B \equiv \int_0^h u dy$ and $A \equiv \int_0^h u^2 dy$, a short exercise in algebra shows $B = U(h - \delta^*)$ and $A = U^2(h - \delta^* - \theta)$ (both follow directly from substituting $u = U - U(1 - u/U)$ into the definitions and using the definitions of δ^*, θ above). Differentiating both with respect to x (product rule, since both $U(x)$ and $\delta^*(x), \theta(x)$ vary) and substituting into $dA/dx - U dB/dx = h\rho^{-1} \cdot$ (RHS pressure term) derived above, every term proportional to the fixed height h cancels exactly (as it must, since h was an arbitrary choice and cannot appear in the final, physical result), leaving, after dividing through by ρU^2 , the **momentum integral (von Kármán) equation**:

$$\boxed{\frac{\tau_w}{\rho U^2} = \frac{d\theta}{dx} + (2\theta + \delta^*) \frac{1}{U} \frac{dU}{dx} .}$$

This is exactly the equation quoted (but not derived) in earlier drafts of this section, now built up honestly from a momentum balance, in precisely the same spirit as every other integral relation in this handout. Given an assumed shape for the velocity profile $u/U = g(y/\delta)$ (satisfying, at minimum, $g(0) = 0$ and $g(1) = 1$), one can compute δ^*, θ and τ_w all as known multiples of $\delta(x)$, substitute into this single ODE, and solve for $\delta(x)$ directly, without ever touching the original PDE. This approximate strategy is called the **Kármán–Pohlhausen method**, and we use it below on the flat plate, where we can check it directly against the exact Blasius numbers just derived.

13.5 Flat-Plate Boundary Layer, by the Approximate (Integral) Method

Take the simplest reasonable profile satisfying the no-slip condition, matching the free stream, and matching the physically required zero shear at the outer edge of the layer ($\partial u / \partial y = 0$ at $y = \delta$, since the shear must vanish smoothly into the shear-free outer flow):

$$\frac{u}{U} = 2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2 \equiv 2\eta - \eta^2, \quad \eta \equiv y/\delta.$$

Compute the three ingredients the momentum integral equation needs, each a clean, elementary integral:

$$\delta^* = \delta \int_0^1 (1 - 2\eta + \eta^2) d\eta = \frac{\delta}{3}, \quad \theta = \delta \int_0^1 (2\eta - \eta^2)(1 - 2\eta + \eta^2) d\eta = \frac{2\delta}{15},$$

$$\tau_w = \mu \frac{\partial u}{\partial y} \Big|_{y=0} = \frac{\mu U}{\delta} (2 - 2\eta) \Big|_{\eta=0} = \frac{2\mu U}{\delta}.$$

For a flat plate $dU/dx = 0$, so the momentum integral equation collapses to $\tau_w/(\rho U^2) = d\theta/dx$. Substituting the results above,

$$\frac{2\mu U/\delta}{\rho U^2} = \frac{2}{15} \frac{d\delta}{dx} \quad \implies \quad \frac{15\nu}{U} = \delta \frac{d\delta}{dx}.$$

This separates immediately: $\delta d\delta = (15\nu/U) dx$, and integrating from the leading edge ($\delta = 0$ at $x = 0$) gives $\delta^2/2 = 15\nu x/U$, or

$$\delta(x) = \sqrt{30} \sqrt{\frac{\nu x}{U}} \approx 5.48 \frac{x}{\sqrt{\text{Re}_x}}.$$

Compare this to the exact Blasius result, $\delta \approx 5.0 x/\sqrt{\text{Re}_x}$, from the last section: the simple parabolic-profile approximation is within about 10%, a remarkably good result from an assumed profile and a handful of elementary integrals, and exactly the kind of trade-off (a modest loss of accuracy for an enormous gain in tractability, since this method works equally well once $dU/dx \neq 0$, where no Blasius-type exact solution exists at all) that makes the integral method so valuable in practice. Following through to the skin friction,

$$c_f = \frac{\tau_w}{\frac{1}{2}\rho U^2} = \frac{4\mu/(\rho U \delta)}{1} = \frac{4\nu}{U\delta} = \frac{4}{\sqrt{30}} \frac{1}{\sqrt{\text{Re}_x}} \approx \frac{0.730}{\sqrt{\text{Re}_x}},$$

again close to the exact $0.664/\sqrt{\text{Re}_x}$. Integrating τ_w over the whole plate (length L , width b) gives the total **skin friction drag**, $D_f = b \int_0^L \tau_w dx$, which for a body like a flat plate, streamlined body or thin wing (where the flow does not separate) is essentially the entire drag; this is called **friction drag**, to distinguish it from the **pressure drag** discussed next. As the local Reynolds number Re_x grows large enough (empirically, $\text{Re}_x \gtrsim 5 \times 10^5$ for a smooth flat plate), the laminar boundary layer becomes unstable, in the same way the whole pipe flow did in Section 12.1, and transitions to turbulent, after which the far more effective turbulent mixing (Section 12.6) makes the boundary layer thicker but also raises τ_w and c_f substantially compared to the laminar values (a turbulent, one-seventh-power assumed profile carried through the same integral method gives the well-known result $c_f \approx 0.0592/\text{Re}_x^{1/5}$, by exactly the same procedure just demonstrated, only with a different assumed shape for u/U).

13.6 Boundary Layers with Pressure Gradients: Separation and Pressure Drag

Up to now we assumed U_∞ , and hence the outer pressure, is constant along the surface (a flat plate). Over a curved body such as a cylinder or an airfoil, the outer inviscid flow speeds up and slows down as it goes around the shape, and by Bernoulli's equation (Section 8.4) the pressure correspondingly falls and rises. This pressure variation is imposed on the boundary layer from outside, and it profoundly changes its behaviour.

In a **favourable** pressure gradient ($dp/dx < 0$, flow accelerating, as on the front half of a cylinder), the pressure is pushing the fluid forward, helping it overcome the retarding effect of wall friction, and the boundary layer stays thin and healthy. In an **adverse** pressure gradient ($dp/dx > 0$, flow decelerating, as on the rear half of a cylinder), the pressure is now pushing the fluid *backward*. The fluid in the outer part of the boundary layer, still carrying plenty of momentum, can fight through this adverse pressure, but the fluid right near the wall has already been slowed down by friction (Section 12.2) and has very little momentum left to spend. If the adverse pressure gradient persists for long enough, the near-wall fluid is brought to a complete stop and then reversed, and the boundary layer **separates** from the surface, forming a large, low-pressure wake behind the body. The separation point is precisely where $\partial u/\partial y|_{y=0} = 0$, i.e. where the wall shear stress first vanishes.

Separation matters enormously because the pressure inside the wake stays close to the (low) pressure at the point of separation, rather than recovering back up to the (higher) stagnation-like pressure that inviscid theory would predict at the back of the body. The resulting fore-aft pressure imbalance produces **pressure drag** (or form drag), which for a bluff body like a sphere or a cylinder typically dwarfs the friction drag of Section 13.5. This is precisely why streamlined shapes, which are designed to keep the pressure gradient favourable (or only very gently adverse) for as long as possible and delay separation, have so much lower drag than bluff bodies of the same frontal area, and it is also why (somewhat counter-intuitively) tripping a boundary layer into turbulence early, as dimples do on a golf ball, can actually *reduce* total drag: a turbulent boundary layer mixes momentum from the outer flow down towards the wall so effectively that it can resist a given adverse pressure gradient much longer than a laminar layer can, delaying separation and shrinking the low-pressure wake by more than enough to offset its own higher friction.

13.7 Application: Thin Airfoil Theory

Away from the thin boundary layer, we argued in Section 9.5 that the flow can often be treated as inviscid and irrotational, governed by Laplace's equation for the velocity potential, $\nabla^2\phi = 0$. **Thin airfoil theory** exploits this, together with the superposition machinery of Section 10, to actually compute the lift of a cambered airfoil from first principles, rather than just asserting the Kutta–Joukowski result as we did in Section 10.4.

Setting up the problem. Represent a thin airfoil of chord c by a sheet of vorticity of strength $\gamma(x)$ (a continuous distribution of the point vortices of Section 10.1) spread along the chord line itself, $0 \leq x \leq c$, rather than around the true, curved surface, an approximation valid precisely because the airfoil is thin. By the same Biot–Savart reasoning that gave the point vortex its velocity field $u_\theta = \Gamma/2\pi r$ in Section 10.1, a short element of sheet $\gamma(\xi) d\xi$ located at ξ induces, at another point x on the chord, a velocity component perpendicular to the chord of

$$dw = -\frac{\gamma(\xi) d\xi}{2\pi(x - \xi)} \quad \implies \quad w(x) = -\frac{1}{2\pi} \int_0^c \frac{\gamma(\xi)}{x - \xi} d\xi$$

(a principal-value integral, since the integrand is singular at $\xi = x$). The airfoil surface, at small angle of attack α and small camber-line slope dy_c/dx , must be a streamline: the total velocity normal to the chord, from the free stream's own small normal component $U_\infty(dy_c/dx - \alpha)$ (the free stream, tilted by α , sees the sloped camber line as if it had an effective normal component; a short exercise in resolving U_∞ into chord-aligned and chord-normal components at small angles gives exactly this) plus the induced velocity $w(x)$, must vanish. This gives the **fundamental equation of thin airfoil theory**:

$$\frac{1}{2\pi} \int_0^c \frac{\gamma(\xi)}{x - \xi} d\xi = U_\infty \left[\alpha - \frac{dy_c}{dx} \right].$$

This is a single integral equation for the one unknown function $\gamma(x)$, given the airfoil's shape $y_c(x)$ and flight condition α ; once we have γ , we already know from Section 10.4 that the total circulation $\Gamma = \int_0^c \gamma dx$ gives the lift directly via Kutta–Joukowski, $L' = \rho U_\infty \Gamma$.

13.8 Glauert's Solution by Fourier Series

Glauert's method solves the integral equation by first changing variables to remove the endpoints' singular behaviour, via the substitution

$$x = \frac{c}{2}(1 - \cos \theta), \quad 0 \leq \theta \leq \pi,$$

which stretches the leading and trailing edges ($x = 0, c$) out to $\theta = 0, \pi$ while compressing the middle of the chord, exactly the kind of change of variables that regularizes an integrable endpoint singularity in γ . Expand $\gamma(\theta)$ in the specific Fourier series

$$\gamma(\theta) = 2U_\infty \left[A_0 \cot \frac{\theta}{2} + \sum_{n=1}^{\infty} A_n \sin n\theta \right],$$

where the leading, cotangent term is chosen deliberately: it is the one distribution (a "fundamental" solution of the airfoil integral equation, analogous to choosing $\sin, \cos$ as the fundamental solutions of a vibrating string) known to produce a *uniform* induced velocity $w = -U_\infty A_0$ along the whole chord, exactly what a uniform angle-of-attack contribution requires, while each $\sin n\theta$ term produces a induced-velocity contribution $U_\infty A_n \cos n\theta$, matching the varying camber slope term by term. Substituting this series into the fundamental equation and using the classical **Glauert integral**,

$$\frac{1}{\pi} \int_0^\pi \frac{\cos n\theta}{\cos \theta - \cos \theta_0} d\theta = \frac{\sin n\theta_0}{\sin \theta_0} \quad (n = 0, 1, 2, \dots),$$

a standard result of Fourier analysis (it follows from expanding $1/(\cos \theta - \cos \theta_0)$ in the same cosine series and using orthogonality, exactly the same toolkit used to solve Laplace's equation by separation of variables), every term in the integral collapses cleanly, and the fundamental equation reduces to a simple statement with no integral left at all:

$$\alpha - \frac{dy_c}{dx}(\theta_0) = A_0 - \sum_{n=1}^{\infty} A_n \cos n\theta_0.$$

Since this must hold at every θ_0 , we can read off the Fourier coefficients directly, exactly as we would find the coefficients of any Fourier series, by comparing this equation to the standard cosine

expansion of the known function $\alpha - dy_c/dx$:

$$A_0 = \alpha - \frac{1}{\pi} \int_0^\pi \frac{dy_c}{dx} d\theta, \quad A_n = \frac{2}{\pi} \int_0^\pi \frac{dy_c}{dx} \cos n\theta d\theta \quad (n \geq 1).$$

Every coefficient is now an explicit, computable integral of the known camber-line slope; the whole problem has been solved analytically, in closed form, for *any* thin camber line, a remarkable payoff for the Fourier-series trick.

13.9 Lift and Moment

The total circulation follows from integrating the vorticity sheet, converting to θ via $dx = (c/2) \sin \theta d\theta$:

$$\Gamma = \int_0^c \gamma dx = \frac{c}{2} \int_0^\pi \gamma(\theta) \sin \theta d\theta = U_\infty c \int_0^\pi \left[A_0 \cot \frac{\theta}{2} \sin \theta + \sum_n A_n \sin n\theta \sin \theta \right] d\theta.$$

Using $\cot(\theta/2) \sin \theta = 1 + \cos \theta$ (a half-angle identity) and the orthogonality of $\sin n\theta$ and $\sin \theta$ (which kills every term except $n = 1$, exactly the same orthogonality that made the Fourier coefficients above so clean), the integral evaluates to $\Gamma = U_\infty c \pi (A_0 + \frac{1}{2} A_1)$. Then, by Kutta–Joukowski (Section 10.4),

$$L' = \rho U_\infty \Gamma = \pi \rho U_\infty^2 c \left(A_0 + \frac{A_1}{2} \right) \quad \Rightarrow \quad C_l \equiv \frac{L'}{\frac{1}{2} \rho U_\infty^2 c} = 2\pi \left(A_0 + \frac{A_1}{2} \right).$$

Substituting the formula for A_0 from the last section and collecting terms,

$$C_l = 2\pi(\alpha - \alpha_{L=0}), \quad \alpha_{L=0} = -\frac{1}{\pi} \int_0^\pi \frac{dy_c}{dx} (1 - \cos \theta) d\theta,$$

where $\alpha_{L=0}$, the **zero-lift angle**, is a property of the camber shape alone (it does not depend on α , since it comes entirely from the A_1 and camber-integral terms) and is negative for any airfoil cambered the "normal" way (curving up towards the trailing edge), meaning such an airfoil still produces some lift even at zero geometric angle of attack. For a **symmetric** airfoil, $dy_c/dx \equiv 0$ everywhere, so every integral vanishes, $\alpha_{L=0} = 0$, and we recover exactly the clean result quoted (without derivation) earlier: $C_l = 2\pi\alpha$.

A similar calculation (multiply the fundamental equation by $\cos \theta_0$ before integrating, or equivalently take the pitching moment of the pressure difference $\Delta p \propto \gamma(x)$ about the leading edge,

weighted by the moment arm x) gives the moment coefficient about the leading edge in terms of the same Fourier coefficients,

$$C_{m,LE} = -\frac{\pi}{2} \left(A_0 + A_1 - \frac{A_2}{2} \right),$$

and transferring this moment to the quarter-chord point ($x = c/4$, using $C_{m,c/4} = C_{m,LE} + C_l/4$ from a simple moment-transfer, exactly the parallel-axis idea we already used for hydrostatic forces in Section 6.4) gives the beautifully simple result

$$C_{m,c/4} = -\frac{\pi}{4}(A_1 - A_2),$$

which depends only on the *camber* (through A_1, A_2) and not at all on α (which enters C_l only through A_0 , absent here). This is the famous statement that the quarter-chord point is the **aero-dynamic center** of a thin airfoil: the point about which the pitching moment is independent of angle of attack, a direct, first-principles consequence of the Fourier decomposition, and a cornerstone of how aircraft longitudinal stability is analyzed.

Thin airfoil theory predicts the lift-curve slope $dC_l/d\alpha = 2\pi$ remarkably well against experiment at small α , but it says nothing at all about **stall**: the theory assumes the vortex sheet stays neatly attached to the chord, exactly the attached, non-separated boundary layer picture of Section 13.6, and breaks down completely once the real boundary layer separates at large α , producing the abrupt loss of lift that no amount of further inviscid analysis can capture, precisely because stall is, at its root, a viscous phenomenon.

13.10 Application: Turbomachinery

Finally, the same tools built throughout this whole handout come together in the analysis of turbomachines, devices such as pumps, compressors, and turbines that add energy to a fluid (via a rotating shaft doing work on it) or extract energy from a fluid (spinning a shaft with it). Applying the angular momentum form of the momentum equation from Section 8.3 to a control volume that rotates with the blades of a pump or turbine, in exactly the same spirit as our pipe bend example,

gives the **Euler turbomachine equation**,

$$\dot{W}_{shaft} = \dot{m}(U_2 V_{t2} - U_1 V_{t1}),$$

where U is the blade speed and V_t is the tangential component of the fluid's absolute velocity at inlet (1) and outlet (2). This equation is precisely the shaft work term $\dot{W}_{shaft}$ that we first introduced, without derivation, in the control volume energy equation of Section 8.4, and it says that the work exchanged is set entirely by how much the blades manage to change the swirl (tangential velocity) of the fluid as it passes through, exactly analogous to how a figure skater's angular momentum changes when they extend or pull in their arms. The velocity triangles used to design real turbomachine blades (breaking the absolute velocity into a blade-relative component and a blade-speed component) are simply a bookkeeping device for evaluating V_{t1} and V_{t2} from the blade geometry and rotational speed, and the boundary layer separation ideas of Section 13.6 govern exactly how much of this ideal, inviscid work prediction is actually achievable before losses (again, ultimately, viscous dissipation, all the way back to Φ in Section 9.4) eat into it. We will take this equation much further, into real pump and turbine performance, in the final chapter.

14 Compressible Flow

Throughout this handout we have leaned on one simplifying assumption more than any other: constant density. It is time to ask what changes when it is not true, because at high enough speeds every gas flow, from a bicycle pump to a rocket nozzle, must be treated as compressible.

14.1 The Speed of Sound

Sound is nothing but a very weak pressure disturbance propagating through a fluid, and we can find how fast it travels with a control volume argument in the same spirit as Section 8. Consider a small pressure pulse moving at speed c (relative to the still fluid ahead of it) into a gas at rest, with pressure p , density ρ . Switch to a frame of reference moving with the wave, so that in this frame gas approaches the (now stationary) wave at speed c , and leaves at a slightly different speed $c - dV$, with a slightly higher pressure $p + dp$ and density $\rho + d\rho$ behind it (a compression wave).

Apply continuity (Section 8.2) to a control volume straddling the thin wave, of cross-section A :

$$\rho A c = (\rho + d\rho) A (c - dV) \implies \rho dV = c d\rho \quad (\text{dropping the tiny product } d\rho dV).$$

Now apply the momentum equation (Section 8.3) to the same control volume, with pressure the only force (the wave is thin, so friction and gravity are negligible over its tiny thickness):

$$pA - (p + dp)A = \dot{m}[(c - dV) - c] = -\rho A c dV \implies dp = \rho c dV.$$

Eliminating dV between these two relations gives $dp = \rho c \left(\frac{c d\rho}{\rho} \right) = c^2 d\rho$, so

$$c^2 = \frac{dp}{d\rho}.$$

The last physical question is: at fixed density, does the pressure respond to a compression fast enough for heat to have time to escape, or is the compression too rapid for any heat transfer to occur? A sound wave oscillates far too quickly for heat conduction to keep up, so the process is essentially adiabatic and, since it is also so weak as to be reversible, isentropic. So we should evaluate $dp/d\rho$ along an isentropic path, and for an ideal gas the isentropic relation from Section 2, $p/\rho^\gamma = \text{const}$, gives $dp/d\rho = \gamma p/\rho = \gamma RT$, using the ideal gas law. Hence

$$c = \sqrt{\gamma RT}.$$

Sound therefore travels faster in a lighter gas (smaller molar mass, larger R) and faster in a hotter gas, and it is a purely local, thermodynamic property of the fluid, not a property of the wave itself.

14.2 The Mach Number and Its Physical Meaning

Define $M = V/c$, the ratio of the local flow speed to the local speed of sound, exactly the dimensionless group we anticipated in Section 10.5. Its physical importance comes from the fact that M controls whether a fluid particle can "warn" the fluid ahead of it that it is coming. A moving body continuously emits weak pressure disturbances (sound waves) that spread outward at speed c relative to the local fluid. If $M < 1$ (**subsonic**), these disturbances always outrun the body and the upstream fluid feels it coming, gently accelerating and parting well in advance, exactly the

smooth streamline patterns we assumed throughout Sections 8-11. If $M > 1$ (**supersonic**), the body outruns its own warning signals entirely; the upstream fluid has no idea anything is coming until the body (or, at a finite distance, the cone of overlapping disturbances known as the **Mach cone**) arrives abruptly, which is exactly why supersonic flow is dominated by sudden, discontinuous changes, the shock waves of Section 14.4, rather than the gentle, gradual variations of subsonic flow. As a rule of thumb, compressibility effects (density changes large enough to matter) become significant once M exceeds about 0.3, exactly the number we flagged without proof back in Section 10.5.

14.3 Stagnation Properties and Isentropic Flow

Imagine bringing a moving gas stream to rest *isentropically* (adiabatically and reversibly), for example at the nose of a probe facing directly into the flow. The resulting state is called the **stagnation state**, denoted with a subscript 0, and it is an enormously convenient reference because, for adiabatic flow with no shaft work, the energy equation of Section 8.4 tells us $h_0 = h + V^2/2$ is constant along the whole flow, whether or not the flow itself is isentropic. For a calorically perfect gas, $h = c_p T$, so

$$T_0 = T + \frac{V^2}{2c_p}.$$

Using $c_p = \gamma R/(\gamma - 1)$ and $c = \sqrt{\gamma RT}$, a short substitution (left as an exercise, exactly the algebra you would use to eliminate c_p and V in favour of $M = V/c$) turns this into the compact, purely-in-terms-of- M form

$$\frac{T_0}{T} = 1 + \frac{\gamma - 1}{2} M^2.$$

If, in addition, the flow between the local point and the stagnation point is isentropic (no losses, so we may use the isentropic relation from Section 2 again), then

$$\frac{p_0}{p} = \left(\frac{T_0}{T}\right)^{\gamma/(\gamma-1)} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{\gamma/(\gamma-1)}, \quad \frac{\rho_0}{\rho} = \left(1 + \frac{\gamma - 1}{2} M^2\right)^{1/(\gamma-1)}.$$

These three relations are the workhorses of compressible flow. Notice that T_0 requires only adiabatic flow (no heat transfer) to be constant along a streamline, and is therefore constant across even a lossy device like a shock, while p_0 and ρ_0 additionally require isentropic (lossless) flow, and *do* drop across a shock, exactly the signature we will use in Section 14.4 to measure how much a shock has

degraded the flow.

14.4 Effect of Area Change: Nozzles and Diffusers

Why does a converging duct speed up a subsonic flow but a converging duct *slow down* a supersonic one? Combine differential continuity, $d(\rho AV) = 0 \Rightarrow d\rho/\rho + dA/A + dV/V = 0$, with the differential form of the (frictionless, adiabatic) momentum equation along a streamtube, $dp = -\rho V dV$ (the compressible cousin of Bernoulli, from the Euler equation of Section 9.3 with $\mu = 0$), and the speed of sound relation $dp = c^2 d\rho$ from Section 14.1 to eliminate $d\rho = dp/c^2 = -\rho V dV/c^2$. Substituting into continuity,

$$-\frac{V dV}{c^2} + \frac{dA}{A} + \frac{dV}{V} = 0 \quad \Rightarrow \quad \frac{dA}{A} = \frac{dV}{V} \left(\frac{V^2}{c^2} - 1 \right) = (M^2 - 1) \frac{dV}{V}.$$

This compact result, the **area-velocity relation**, contains the entire qualitative behaviour of compressible nozzles and diffusers. For $M < 1$, the factor $(M^2 - 1)$ is negative, so dA and dV have opposite signs: a converging duct ($dA < 0$) accelerates the flow ($dV > 0$), exactly the incompressible intuition of Section 8.2 baked in as a special case. But for $M > 1$, $(M^2 - 1)$ is positive, so dA and dV have the *same* sign: a converging duct now *decelerates* a supersonic flow, and a diverging duct accelerates it, the complete opposite of low-speed intuition, because at supersonic speed the density is falling so fast (as computed above) that the area must actually increase to keep accommodating the same mass flow at an ever-increasing velocity. Finally, at $M = 1$ exactly, $dA = 0$ regardless of dV : sonic conditions can only occur at a minimum in area, a **throat**.

This is precisely why a rocket or a supersonic wind tunnel uses a **converging-diverging (de Laval) nozzle**: gas is accelerated subsonically through the converging section, reaches exactly $M = 1$ at the throat (the one place a smooth transition through the sonic point is geometrically allowed), and then continues to accelerate supersonically through the diverging section. If the throat reaches $M = 1$, the nozzle is said to be **choked**: the mass flow rate has reached its absolute maximum for the given upstream stagnation conditions, and no further reduction in downstream (back) pressure can push any more mass through, because (exactly as in Section 14.2) no pressure signal from downstream can travel upstream past a sonic or supersonic throat to "tell" the reservoir to supply more.

14.5 Normal Shock Waves

A **normal shock** is an extremely thin region (a few molecular mean free paths) across which a supersonic flow abruptly and irreversibly becomes subsonic, with sudden jumps in pressure, density and temperature; it is the compressible-flow analogue of the hydraulic jump we will meet in the very next chapter, and both are, at heart, the same phenomenon: a sudden transition from a "fast, shallow-energy" state to a "slow, high-energy" state, forced by the requirement that mass, momentum and energy all still balance across a region where friction is completely negligible everywhere except right inside the thin transition itself.

Take a stationary control volume straddling the shock, with upstream (supersonic) state 1 and downstream state 2. Since the shock is thin, we can neglect area change, gravity and heat transfer across it, and continuity, momentum and energy (Sections 8.2 through 8.4) give

$$\rho_1 V_1 = \rho_2 V_2, \quad p_1 + \rho_1 V_1^2 = p_2 + \rho_2 V_2^2, \quad h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}.$$

These are three equations relating the two states, and rather than grinding through the full elimination by brute force, there is an elegant shortcut, **Prandtl's relation**, that gets us most of the way there. Define the **critical speed of sound** a^* as the speed of sound at the (hypothetical) point where the flow, brought isentropically to $M = 1$, would have $V = a = a^*$; the energy equation, written using $a^2 = \gamma RT$ and $h = c_p T = a^2/(\gamma-1)$, becomes $a^2/(\gamma-1) + V^2/2 = a^{*2}(\gamma+1)/[2(\gamma-1)]$ (a constant, the same h_0 at every state, evaluated once at the sonic point), which rearranges to

$$a^2 = \frac{(\gamma+1)a^{*2} - (\gamma-1)V^2}{2}.$$

Now substitute $\rho = p/(RT) = \gamma p/a^2$ into the momentum equation $p_1 + \rho_1 V_1^2 = p_2 + \rho_2 V_2^2$, divide through by the mass flux $G = \rho_1 V_1 = \rho_2 V_2$ (continuity), and use $p/\rho = a^2/\gamma$:

$$\frac{a_1^2}{\gamma V_1} + V_1 = \frac{a_2^2}{\gamma V_2} + V_2 \quad \implies \quad \frac{1}{\gamma} \left(\frac{a_1^2}{V_1} - \frac{a_2^2}{V_2} \right) = V_2 - V_1.$$

Substitute the boxed expression for a_1^2 and a_2^2 above into the left side; the a^{*2} terms combine over a common factor of $(V_2 - V_1)/(V_1 V_2)$ and the V^2 terms combine over a factor of $(V_2 - V_1)$ (a short

but entirely mechanical simplification), leaving

$$\frac{V_2 - V_1}{\gamma} \left[\frac{(\gamma + 1)a^{*2}}{2V_1V_2} + \frac{\gamma - 1}{2} \right] = V_2 - V_1.$$

Since a genuine shock has $V_1 \neq V_2$, we may divide out the common factor $(V_2 - V_1)$, and solving the remaining bracket for a^{*2} gives the beautifully clean **Prandtl relation**:

$$\boxed{V_1V_2 = a^{*2}}.$$

Defining $M^* \equiv V/a^*$, this says simply $M_1^*M_2^* = 1$: since M^* increases monotonically with the ordinary Mach number M (they agree at $M = 1$ and move together), $M_1^* > 1$ forces $M_2^* < 1$, confirming algebraically what physical reasoning already told us, a shock always takes a supersonic upstream state to a subsonic downstream one. From here, expressing a^{*2} in terms of M_1 (using the boxed a^2 formula once more, evaluated at station 1) and substituting back through continuity and the ideal gas law is now just bookkeeping, and carrying it through (an exercise in careful but mechanical substitution, not in any new physics) yields the classic normal shock relations purely in terms of the upstream Mach number M_1 :

$$M_2^2 = \frac{1 + \frac{\gamma-1}{2}M_1^2}{\gamma M_1^2 - \frac{\gamma-1}{2}}, \quad \frac{p_2}{p_1} = \frac{2\gamma M_1^2 - (\gamma - 1)}{\gamma + 1}, \quad \frac{\rho_2}{\rho_1} = \frac{V_1}{V_2} = \frac{(\gamma + 1)M_1^2}{(\gamma - 1)M_1^2 + 2}.$$

The energy equation (which involves only $h_0 = h + V^2/2$) shows $T_{0,1} = T_{0,2}$: stagnation temperature is unchanged, exactly as we noted in Section 14.3, since the shock, though highly irreversible, is still adiabatic. But solving the full set of equations shows entropy strictly increases, $s_2 > s_1$, whenever $M_1 > 1$, and correspondingly $p_{0,2} < p_{0,1}$: the stagnation pressure, our clean marker of "how much useful energy the flow still has to give," is irreversibly lost across the shock, in direct analogy to the head loss h_L we introduced for pipe friction in Section 12.4. If one instead tries to solve these same three equations starting from a *subsonic* $M_1 < 1$, mathematics permits a solution, but it always predicts $s_2 < s_1$, a decrease in entropy with no compensating heat rejection: a violation of the second law. This is why normal shocks are only observed going from supersonic to subsonic, never the reverse. It is a beautiful, concrete illustration of an idea that has been implicit throughout this whole handout, all the way back to the entropy generation term in Section 2's isentropic-flow discussion: the direction of every real, irreversible process in this course, whether it is pipe friction,

boundary layer separation, or a shock wave, is fixed by the same second law of thermodynamics.

15 Open Channel Flow

Every flow we have analyzed since Section 12 has been driven through a closed, completely full conduit, with pressure available at every point to push the fluid along. An **open channel** flow (a river, an irrigation canal, a partially filled storm drain) is fundamentally different: the fluid has a free surface exposed to constant atmospheric pressure, and it is gravity, acting on the slope of the channel bed, rather than an imposed pressure difference, that drives the flow forward. This single change in the driving mechanism is responsible for essentially all of the new phenomena in this chapter.

15.1 Specific Energy and Critical Depth

Consider a wide, rectangular channel carrying a flow rate per unit width $q = Q/b$, at a local depth y and mean velocity $V = q/y$ (continuity, exactly as in Section 8.2, just with the channel width playing the role of a fixed pipe area). Take the channel bottom as the elevation datum and define the **specific energy** as the mechanical energy per unit weight, measured relative to the channel bed rather than some fixed reference:

$$E = y + \frac{V^2}{2g} = y + \frac{q^2}{2gy^2}.$$

This is nothing but Bernoulli's equation from Section 8.4, with atmospheric pressure at the free surface making the pressure term drop out entirely and the elevation z replaced by the local depth y . For a fixed q , $E(y)$ has a very distinctive shape: as $y \rightarrow \infty$, $E \rightarrow y$ (energy dominated by elevation, in the limit of a deep, slow-moving channel); as $y \rightarrow 0$, $E \rightarrow q^2/2gy^2 \rightarrow \infty$ (a thin, fast sheet of water carries enormous kinetic energy). Somewhere in between, $E(y)$ must have a minimum, and setting $dE/dy = 0$,

$$\frac{dE}{dy} = 1 - \frac{q^2}{gy^3} = 0 \quad \implies \quad y_c = \left(\frac{q^2}{g} \right)^{1/3},$$

called the **critical depth**. Substituting back, the corresponding velocity satisfies $V_c^2 = q^2/y_c^2 = gy_c$, or

$$Fr \equiv \frac{V}{\sqrt{gy}} = 1 \quad \text{at } y = y_c,$$

exactly the Froude number we defined by dimensional analysis in Section 10.5, and it has an equally clean physical meaning as the Mach number of Section 14.2: $\sqrt{gy}$ is precisely the speed at which a small surface gravity wave propagates in shallow water of depth y (a fact you can derive with the exact same traveling-disturbance control volume argument we used for the speed of sound), so Fr measures the ratio of the flow speed to the speed at which surface disturbances can travel.

For a given specific energy E (below the value needed to reach y_c , there is no solution at all: the channel cannot carry that much flow at that little energy), there are generally two depths that share the same E , one deep and slow, with $y > y_c$ and $Fr < 1$ (**subcritical** flow, exactly analogous to subsonic flow: surface disturbances can propagate upstream, and the flow "knows" what is coming), and one shallow and fast, with $y < y_c$ and $Fr > 1$ (**supercritical** flow, the open-channel analogue of supersonic flow: disturbances are swept downstream and cannot work their way back up against the current). These two depths sharing the same specific energy are called **alternate depths**.

15.2 Uniform Flow: the Chezy and Manning Equations

In a long, straight channel of constant slope and cross-section, the flow eventually settles into **uniform flow**, where the depth no longer changes along the channel, called the **normal depth**. Physically, this is a balance, not an acceleration: apply the momentum equation (Section 8.3) to a control volume of length L along the channel bed, which slopes downward at a small angle so that $\sin \theta \approx S_0$ (the bed slope). The two forces along the flow direction are the downslope component of the weight, $\rho g A L S_0$ (where A is the cross-sectional flow area), and the retarding wall shear stress τ_w acting over the wetted perimeter P over the length L . For uniform flow the momentum flux in equals the momentum flux out exactly (the velocity profile is unchanging along x , precisely the "fully developed" idea of Section 12.3 transplanted to a free surface), so these two forces must balance directly:

$$\rho g A L S_0 = \tau_w P L \quad \implies \quad \tau_w = \rho g R_h S_0, \quad R_h \equiv \frac{A}{P},$$

where R_h , the **hydraulic radius**, is the open-channel stand-in for the pipe diameter of Section 12: it is the flow area per unit of wetted, friction-generating perimeter (for a full circular pipe, $R_h = D/4$,

exactly one quarter of the diameter, a conversion worth remembering when carrying pipe-flow correlations over to a channel). Modelling the wall shear the same way we modelled turbulent pipe friction in Section 12.6 (writing $\tau_w = \rho C_f V^2/2$ using some empirical friction coefficient C_f , rather than solving the closure problem from scratch) and solving for V gives the **Chezy equation**, $V = C\sqrt{R_h S_0}$, and the most common empirical form for C , valid for the fully rough turbulent flow typical of natural channels, is the **Manning equation**,

$$V = \frac{1}{n} R_h^{2/3} S_0^{1/2} \quad (\text{SI units}),$$

where n is the empirical **Manning roughness coefficient**, playing exactly the role that the relative roughness ε/D played on the Moody chart of Section 12.7, just packaged for a channel instead of a pipe. Given a channel shape, slope and roughness, this single equation, combined with $Q = VA$ (and the fact that both A and R_h depend on the unknown depth y through the channel geometry), can be solved (usually by iteration, since A and R_h are rarely simple functions of y) for the normal depth that will carry a given discharge Q .

15.3 The Hydraulic Jump

Just as a supersonic flow (Section 14.4) cannot smoothly become subsonic and instead does so through an abrupt, dissipative shock, a supercritical open-channel flow ($Fr > 1$) cannot smoothly become subcritical, and instead undergoes an abrupt, turbulent, highly dissipative transition called a **hydraulic jump**: the churning, foamy standing wave you see below a spillway or a sluice gate. We can find the relationship between the depths before (y_1) and after (y_2) the jump with exactly the same control-volume strategy used for the normal shock: take a control volume straddling the (short) jump, and note that although the jump dissipates a great deal of energy (so we *cannot* use Bernoulli's equation, Section 8.4, across it, exactly as we could not use it across a shock), the pressure forces on the two end faces of the control volume are still simple hydrostatic distributions (Section 6.5, since the flow just before and just after the jump is nearly uniform and horizontal), and momentum (Section 8.3, not energy) is still exactly conserved, since the jump generates no external horizontal force other than these two pressure forces. For a wide rectangular channel, the hydrostatic force per unit width at a depth y is $\frac{1}{2}\rho g y^2$ (recall the triangular pressure distribution

and its resultant, from Section 6.5), so

$$\frac{1}{2}\rho g y_1^2 - \frac{1}{2}\rho g y_2^2 = \rho q(V_2 - V_1) = \rho q \left(\frac{q}{y_2} - \frac{q}{y_1} \right).$$

A little algebra (factor $y_1^2 - y_2^2 = (y_1 - y_2)(y_1 + y_2)$ on the left, and clear the fractions on the right) turns this into a clean quadratic in the depth ratio, whose physically meaningful (positive) root is

$$\frac{y_2}{y_1} = \frac{1}{2} \left(\sqrt{1 + 8Fr_1^2} - 1 \right),$$

where $Fr_1 = V_1/\sqrt{gy_1}$ is the (necessarily supercritical, $Fr_1 > 1$) upstream Froude number; you can check that this formula gives $y_2 = y_1$ only in the trivial limit $Fr_1 \rightarrow 1$, and $y_2 > y_1$ for every genuine jump. Just as with the normal shock, one can show that the reverse transition, a subcritical flow jumping up in speed to a faster, shallower supercritical state, would require a *decrease* in mechanical energy production, which cannot happen without a corresponding loss (the jump is dissipative, not energy-generating), so real hydraulic jumps always run in the single direction, supercritical to subcritical, we have derived here; the dissipated energy, $\Delta E = E_1 - E_2$, is exactly what makes the base of a spillway such an effective, deliberately designed place to destroy the kinetic energy of fast-moving water before it can erode the channel further downstream.

16 Turbomachinery

16.1 Velocity Triangles and the Euler Equation Revisited

Recall the Euler turbomachine equation from the end of the last chapter, $\dot{W}_{shaft} = \dot{m}(U_2 V_{t2} - U_1 V_{t1})$, derived from angular momentum conservation (Section 8.5) applied to a rotating blade row. To actually use this equation for design, we need a systematic way to find V_{t1} and V_{t2} , and this is exactly what a **velocity triangle** provides: at any station, the absolute velocity $\mathbf{V}$ seen by a stationary observer is the vector sum of the blade's local velocity $\mathbf{U}$ (purely tangential, $U = \omega r$) and the velocity $\mathbf{W}$ of the fluid *relative to the moving blade* (which is what actually must align with the blade's physical shape, since the fluid has no choice but to follow the blade passage in the rotating frame),

$$\mathbf{V} = \mathbf{U} + \mathbf{W}.$$

For a pump or compressor, the designer chooses the inlet and outlet blade angles so that $\mathbf{W}$ enters and leaves smoothly aligned with the blade, and then works backward through this vector triangle at each station to find V_t , exactly analogous to decomposing velocity into radial and tangential components anywhere else in this handout. The Euler equation then says the ideal (inviscid) work per unit mass is $w = U_2 V_{t2} - U_1 V_{t1}$; for a purely radial-inflow pump with no inlet swirl ($V_{t1} = 0$, a common and efficient design choice, since any swirl brought into the machine has to be paid for without doing useful work), this collapses further to $w = U_2 V_{t2}$, showing directly that the ideal head rise of a pump is controlled entirely by the blade tip speed U_2 and by how tangentially (rather than radially) the fluid is thrown off the impeller.

16.2 Pump Performance: Head, Power, and Efficiency

For a real pump, define the **head** added to the fluid, using the energy equation of Section 8.4 between the pump's inlet and outlet (with the pressure, velocity and elevation terms usually collected together, since a pump is a short device and the difference is what matters),

$$h_p = \frac{p_2 - p_1}{\rho g} + \frac{V_2^2 - V_1^2}{2g} + (z_2 - z_1),$$

so that the fluid (or hydraulic) power delivered is $P_w = \rho g Q h_p$. This is always less than the shaft power $P_{shaft} = \omega T_{shaft}$ actually supplied to the pump, because of three distinct kinds of loss: hydraulic losses (friction and separation inside the impeller and casing, exactly the boundary-layer and pipe-friction losses of Sections 12 and 13, applied to the pump's own internal passages), volumetric losses (some fluid leaks back internally through clearances instead of being usefully pumped forward, a parasitic recirculation that adds no net head), and mechanical losses (bearing and seal friction on the shaft itself, which never even reaches the fluid). All three are lumped into the overall **pump efficiency**, $\eta_p = P_w / P_{shaft}$.

Neither h_p nor η_p is a single fixed number for a given pump; both vary with the flow rate Q the pump is asked to deliver, and a manufacturer's **performance curve** plots exactly this variation, $h_p(Q)$ and $\eta_p(Q)$ at a fixed rotational speed. At shutoff ($Q = 0$, the pump running against a closed valve) h_p is at its maximum but $\eta_p = 0$ (all power wasted as heat, none delivered as useful flow work), efficiency rises to a peak at some **best efficiency point (BEP)**, and h_p typically falls off as Q increases further, since the internal losses (which we would expect, from Section 12, to scale

roughly with $V^2 \propto Q^2$) grow faster than the ideal, loss-free head predicted by the Euler equation. A pump should always be selected to run as close to its BEP as the rest of the system allows.

16.3 The System Curve and the Operating Point

A pump never operates in isolation; it is always pushing fluid through some piping system, and the flow rate that actually results is set by where the pump's own head-flow curve intersects the demand of that system. Write the energy equation (Section 8.4, now including the pump as a source of head and the losses of Section 12.4 as a sink of head) around the whole piping loop the pump serves,

$$h_p = (z_2 - z_1) + \frac{p_2 - p_1}{\rho g} + \sum h_L(Q),$$

where the right-hand side, the **system curve**, is simply the static lift the pump must overcome plus all the friction and minor losses (Sections 12.4 and 12.7) that grow with Q^2 , since $h_L \propto V^2 \propto Q^2$ for turbulent flow at a fixed friction factor. The actual flow delivered is found graphically (or by solving simultaneously) at the intersection of the pump's own $h_p(Q)$ curve from the last section with this system curve, and this single graphical construction, the point where a machine's own characteristic meets the demand placed on it by everything downstream, is exactly the same idea that will determine which point on a fan curve, a compressor map, or (in reverse) a turbine's own characteristic actually gets realized in service.

16.4 Dimensional Analysis of Turbomachines and the Affinity Laws

Just as in Section 10, we can apply the Buckingham II theorem to a pump, treating the head rise gh_p , shaft power P , flow rate Q , impeller diameter D , rotational speed N , and fluid properties ρ, μ as the relevant variables. Carrying through the repeating-variables method (using ρ, N, D as repeats, exactly as we used L, V, ρ for the sphere in Section 10.2) produces three key dimensionless groups, conventionally called the **flow, head, and power coefficients**:

$$C_Q = \frac{Q}{ND^3}, \quad C_H = \frac{gh_p}{N^2 D^2}, \quad C_P = \frac{P}{\rho N^3 D^5}.$$

The Buckingham theorem guarantees that, for a family of geometrically similar pumps (same shape, different size and speed), C_H and C_P are each some universal function of C_Q alone (dropping a

weak dependence on a pump Reynolds number, which is usually justified since pump flows are almost always highly turbulent, exactly the high-Re regime of Section 10.5 where the friction factor itself becomes nearly independent of Re). This has an enormously useful practical consequence, the **pump affinity laws**: for the *same* pump (fixed D) running at two different speeds N_1, N_2 , matching C_Q tells us $Q_2/Q_1 = N_2/N_1$, and then matching C_H and C_P at that same operating point gives

$$\frac{Q_2}{Q_1} = \frac{N_2}{N_1}, \quad \frac{h_{p,2}}{h_{p,1}} = \left(\frac{N_2}{N_1}\right)^2, \quad \frac{P_2}{P_1} = \left(\frac{N_2}{N_1}\right)^3,$$

so that flow scales linearly with speed, head with the square of speed, and power with the cube of speed. This is exactly why a small reduction in a pump or fan's rotational speed (for instance, via a variable-frequency drive) produces a disproportionately large saving in power, a fact of real economic importance in HVAC and municipal pumping systems.

A further, especially useful combination of C_Q and C_H , chosen specifically to eliminate the size D (so that it can be used to compare pumps of *different* sizes and *different* types, not just the same pump at different speeds), is the **specific speed**,

$$N_s = \frac{N\sqrt{Q}}{(gh_p)^{3/4}},$$

evaluated at each pump's own best efficiency point. Because N_s is dimensionless (or expressed in a standard set of mixed units, depending on convention) and independent of the actual size of the machine, it depends only on the pump's *shape*, and it turns out to correlate remarkably well with which broad family of impeller design is most efficient for a given duty: low N_s favours a narrow, purely radial-flow (centrifugal) impeller, suited to high head and low flow, while high N_s favours a wide, purely axial-flow impeller (a propeller), suited to low head and high flow, with mixed-flow designs occupying the range in between. This single number is the primary tool engineers use to select which type of pump (or turbine, in the next section) to even consider for a new duty, before any detailed design work begins.

16.5 Cavitation and Net Positive Suction Head

Everything so far has assumed the fluid stays liquid throughout the pump. But recall from the pressure-drop discussion of Bernoulli's equation (Section 8.4) that wherever a flow speeds up, its

pressure must fall; inside a pump impeller, the local pressure can drop very low indeed, especially right at the impeller eye (inlet) or on the low-pressure (suction) side of each blade, precisely where the local velocity is highest. If this local pressure drops all the way to the fluid's vapor pressure p_v at the local temperature, the liquid boils, forming small vapor bubbles; when these bubbles are swept downstream into a region of higher pressure they collapse violently, an implosive process called **cavitation**, which produces noise, vibration, a sharp drop in performance, and, over time, severe pitting erosion of the blade surface, exactly the sort of unwanted, uncontrolled phase change that pump designers work hard to avoid entirely.

To avoid it, we need to guarantee that the pressure everywhere in the pump inlet stays comfortably above p_v . Define the **net positive suction head available**, using the energy equation (Section 8.4) evaluated at the pump inlet flange relative to the vapor pressure,

$$\text{NPSHA} = \frac{p_i}{\rho g} + \frac{V_i^2}{2g} - \frac{p_v}{\rho g},$$

which depends only on the suction-side piping layout (the elevation and losses between the fluid supply and the pump inlet, exactly the same energy-equation bookkeeping as Section 12.4) and not on the pump itself. The pump manufacturer, in turn, publishes the **net positive suction head required**, $\text{NPSHR}(Q)$, an empirically measured property of that specific pump design representing how much of a safety margin above p_v the pump's own internal geometry needs at the inlet before internal pressure drops trigger cavitation. Safe operation requires simply $\text{NPSHA} > \text{NPSHR}$ at the intended operating point, and this single inequality is exactly why pumps handling hot liquids (where p_v is high) or drawing from a source well below the pump (where the elevation term in NPSHA is unfavourable) are so much more prone to cavitation trouble in practice.

16.6 Turbines: Impulse and Reaction Machines

A turbine runs the whole energy exchange of Section 15.1 in reverse: the fluid does work *on* the blades, so the Euler equation becomes $\dot{W}_{shaft} = \dot{m}(U_1 V_{t1} - U_2 V_{t2})$ (extracted, rather than added, swirl), and turbines are broadly classified by *how* the fluid delivers that work.

In an **impulse turbine**, exemplified by the **Pelton wheel** used for very high head, low flow hydropower sites, all of the available head is converted into kinetic energy *before* the fluid ever touches the rotor, through a stationary nozzle (governed entirely by the incompressible Bernoulli equation

of Section 8.4, exactly as in the vena-contracta-free ideal nozzle flow we have used throughout), producing a single high-speed jet at essentially atmospheric pressure. This jet then strikes a series of curved buckets mounted on the wheel; since the pressure is atmospheric on both sides of the bucket, no pressure-based (reaction) force is at play, only the momentum change of the jet as the buckets deflect it, exactly the pipe-bend, momentum-equation logic of Section 8.3, now applied to a jet striking a moving surface instead of fluid confined inside a fixed pipe. A short optimization of the momentum-equation result (maximizing power over bucket speed U , for a jet of speed V_1) shows the ideal bucket speed is exactly $U = V_1/2$, half the jet speed, a clean design target that follows directly from first principles rather than from any empirical correlation.

In a **reaction turbine** such as the **Francis** (mixed-flow, used across a broad range of medium heads) or **Kaplan** (axial-flow, adjustable-pitch, used for very low heads and high flows, such as run-of-river hydropower) designs, by contrast, the fluid remains at an elevated pressure as it passes through the rotor itself, and both a drop in pressure *and* a drop in velocity (through carefully shaped, converging blade passages) contribute to the work extracted, so that the casing must fully enclose and seal the rotor against the surrounding pressure, unlike the impulse turbine's rotor, which simply spins in open air. Exactly the specific-speed classification introduced in Section 15.4 for pumps applies equally well here, now describing turbines instead: Pelton wheels occupy the lowest specific-speed range (very high head, comparatively low flow), Francis turbines the middle range, and Kaplan turbines the highest range (low head, high flow), and this single dimensionless number, derived from nothing more than the Buckingham theorem of Section 10.2 applied to a rotating machine, is exactly what a hydropower engineer consults first when matching a turbine design to a given river's head and flow.

And with that, we have come full circle: from a first law of thermodynamics for a control mass in Section 2, all the way to the selection of a real hydroelectric turbine, every single idea in this handout, conservation of mass, Newton's second law, conservation of energy, and the dimensional reasoning that ties them all into usable numbers, has been applied over and over, to smaller and smaller pieces of fluid, to faster and faster flows, and to larger and larger, more realistic pieces of engineering hardware. If you keep that unifying thread in mind, the rest of your fluid mechanics education, and quite possibly your engineering career, will be a great deal less intimidating.

Figure 5: The Moody chart, generated by solving the Colebrook equation for f at each Re , for a family of relative roughness values ε/D . The laminar line $f = 64/\text{Re}$ is shown in black; the grey band marks the transitional range $2300 < \text{Re} < 4000$, where neither relation reliably applies.

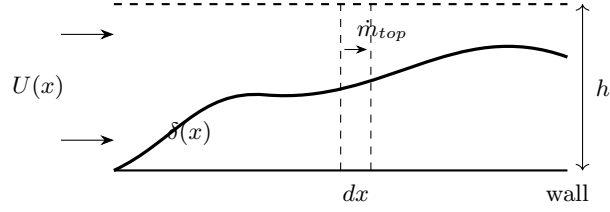


Figure 6: Control volume used to derive the momentum integral equation. The top face, above the growing boundary layer everywhere, admits an inflow of freestream fluid $\dot{m}_{top}$.

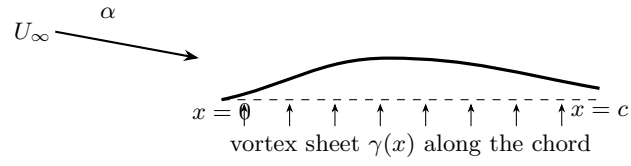


Figure 7: Thin airfoil theory replaces the curved camber line by a vortex sheet of strength $\gamma(x)$ on the chord.